

# Daniel Ari Friedman, PhD

Computational biologist, Active Inference researcher, cognitive security researcher, educator, and artist | California, USA, Earth

|        |                                                                                                              |
|--------|--------------------------------------------------------------------------------------------------------------|
| Verify | <a href="#">resume/verify.html</a><br>QR target, public hashes, source files, and artifact sizes.            |
| Source | <b>manifest sha256 225322fa4052110b</b><br>resume/source.json + canonical data exports                       |
| Output | <b>json sha256 ddb0b93a30f5a4bc</b><br><a href="#">data/resume.json</a><br><a href="#">resume/resume.pdf</a> |



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|-----------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Curated Works</b><br><b>193</b><br><a href="#">data/works.json</a> | <b>Software Rows</b><br><b>95</b><br><a href="#">data/software.json</a>     | <b>Scholar Metrics</b><br><b>777</b><br><a href="#">Scholar profile</a>     | <b>Resume Records</b><br><b>119</b><br><a href="#">verify page</a>          |
| <b>Source Hash</b><br><b>225322fa4052</b><br>source manifest          | <b>JSON Hash</b><br><b>ddb0b93a30f5</b><br><a href="#">data/resume.json</a> | <b>PDF Hash</b><br><b>verify page</b><br><a href="#">resume/verify.html</a> | <b>Footer QR</b><br><b>every page</b><br><a href="#">resume/verify.html</a> |

## Source-to-Output Flow

| Inputs                                                                                                                                        | Generated JSON                                                                    | Public PDF + Verification                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <a href="#">resume/source.json</a><br><a href="#">data/works.json</a><br><a href="#">data/software.json</a><br>source sha256 225322fa4052110b | <a href="#">data/resume.json</a><br>214,470 bytes<br>json sha256 ddb0b93a30f5a4bc | <a href="#">resume/resume.pdf</a><br><a href="#">resume/verify.html</a><br>PDF hash posted on verification page |

## Section Counts

|                  |    |
|------------------|----|
| Education        | 4  |
| Experience       | 16 |
| Awards           | 12 |
| Conferences      | 17 |
| Media / Outreach | 29 |
| Service          | 27 |
| Art Uses         | 14 |



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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| <b>GitHub</b> <a href="https://github.com/docxology">https://github.com/docxology</a>                                                                                              | <b>LinkedIn</b> <a href="https://www.linkedin.com/in/danielarifriedman/">https://www.linkedin.com/in/danielarifriedman/</a>                                        |
| <b>Scholar</b> <a href="https://scholar.google.com/citations?user=DXjPFtYAAAAJ">https://scholar.google.com/citations?user=DXjPFtYAAAAJ</a>                                         | <b>ORCID</b> 0000-0001-6232-9096                                                                                                                                   |
| <b>Keybase / ENS</b> docxology / docxology.eth                                                                                                                                     | <b>Art</b> <a href="https://www.flickr.com/photos/daniel_friedman/">https://www.flickr.com/photos/daniel_friedman/</a>                                             |

|                                                            |                                                        |                                                                          |
|------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Curated Works</b><br><b>193</b><br>from data/works.json | <b>Software Rows</b><br><b>95</b><br>61 owned + 34 All | <b>Scholar Citations</b><br><b>777</b><br>h 15 / i10 17 as of 2026-06-09 |
|------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------|

Variant: full. Generated: 2026-07-12T18:43:12Z. Current public metrics: 193 curated works; 95 catalogued software repositories (61 owned + 34 All); 777 Google Scholar citations, h-index 15, i10-index 17 as of 2026-06-09.

## Summary

Multidisciplinary researcher and open-source builder working across social insect biology, Active Inference, cognitive security, computational frameworks, education, and art.

## Education (4)

### 2019-2023 Postdoctoral Fellow | University of California, Davis | Davis, CA, USA

- Co-mentors: Prof. Brian Johnson (UC Davis) and Prof. Tim Linksvayer (Arizona State University).
- NSF Postdoctoral Research Fellowship in Biology (PRFB), award DBI-2010290; NSF budget period 2020-2022 with affiliation extending to 2023.
- Genomics, epigenomics, transcriptomics, and behavior in eusocial insects, mainly ants and bees.
- Science research, mentoring, participation, service, and organizing events and communities online.

Source: NSF DBI-2010290 <https://grantome.com/grant/NSF/DBI-2010290> | UC Davis <https://www.ucdavis.edu/>

### 2014-2019 PhD | Stanford University | Stanford, CA, USA

- Advisor: Professor Deborah M. Gordon, Biology.
- Research, teaching, service, and coursework from Summer 2014 to Summer 2019.
- Thesis topics: ant ecology, collective behavior, brain gene expression, and neurometabolism.
- Stanford Graduate Fellow (Urbanek Family Fellowship), 2014-2017.
- Co-organizer of the Stanford Complexity Group, 2016-2019.
- Stanford Neuroscience NeuroChoice Grant for neurometabolism in ants, 2019, \$25,000.
- Stanford Systems Biology Seed Grant for gene expression and behavior in ants, 2018, \$25,000.
- Theodore Roosevelt Memorial Award, American Museum of Natural History, 2016, \$2,500.
- Lewis and Clark Fund grant, American Philosophical Society, 2016, \$4,000.
- NSF GRFP Honorable Mention, 2015 and 2016.

Source: Stanford dissertation <http://purl.stanford.edu/pb813wm1484>



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**2010-2014 BSc in Genetics | University of California, Davis | Davis, CA, USA**

- BSc in Genetics, University of California, Davis.
- Genetics Department Citation, Highest Honors, 2014; 3.74 GPA overall.
- Campus Community Service Silver Award, 2014.
- Stanley and Jacqueline Schilling Award for Service, 2014.
- Completed the Population Biology Graduate Group one-year base PhD curriculum, 2013.
- Co-founder of The Aggie Transcript, an undergraduate biology journal, 2014.
- Victoria Finnerty Award for Research in Drosophila, 2013.
- Elected member of Phi Beta Kappa, UC Davis Chapter, 2014.
- Co-organizer of the Chess Club at UC Davis, 2012-2014.
- Research Scholars Program in Insect Biology, 2011-2014.
- Provost's Undergraduate Fellowship for Drosophila research, 2011, \$2,500.

Source: UC Davis <https://www.ucdavis.edu/>

**2006-2010 High School | Crystal Springs Uplands School | Hillsborough, CA, USA**

- Crystal Springs Uplands School.

Source: Crystal Springs Uplands School <https://www.crystal.org/>

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**Experience (16)****2026-ongoing Del Norte Triplicate | Typographical Editor**

Community Organization

- Typographical editing for The Triplicate newspaper of Del Norte County and Crescent City.

Source: The Triplicate <https://www.triplicate.com/>

**2021-ongoing Active Inference Institute | Co-founder, Participant, Officer, President, Treasurer**

Community Organization

- Co-founded the Active Inference Institute; current President and Treasurer.

Source: All officers <https://www.activeinference.institute/officers> | All <https://activeinference.org/>

**2019-2024 Complexity Adventures | Co-founder, Facilitator, Participant, Organizer**

Community Organization

- Organized activities and community around learning and applying complexity science.

Source: Complexity Adventures profile <https://www.complexityadventures.com/organizers/2>

**2019-ongoing COGSEC | Researcher**

Community Organization

- Research facilitation and participation on initiatives related to cognitive security.

Source: COGSEC <https://www.cogsec.org/>

**2014-2019 Stanford Complexity Group | Organizer, Participant**

Community Organization

- Organized the Stanford Complexity Group while it was active.

Source: Stanford <https://www.stanford.edu/> | verification <https://danielarifriedman.com/resume/verify.html>

**2025-2026 Stealth Startup | Operator**

Startup

- Systems modeling and related work to be public later.

Source: verification <https://danielarifriedman.com/resume/verify.html>

**2023-2024 Stealth Startup | Co-founder, Operator**

Startup

- Stealth Active Inference startup.

Source: verification <https://danielarifriedman.com/resume/verify.html>



**2019-2023 Remotor | Co-founder, Researcher**

Startup

- Education, online event planning, and consulting for remote teams.

Source: verification <https://danielarifriedman.com/resume/verify.html>**2017-2019 fm.analytics | Co-founder, Researcher**

Startup

- Predictive analytics and dashboard visualizations with Prof. Glenn Magerman at KU Leuven.

Source: verification <https://danielarifriedman.com/resume/verify.html>**2017-2019 Cryptoptera | Co-founder, Researcher**

Startup

- Three-person cryptocurrency research group.

Source: verification <https://danielarifriedman.com/resume/verify.html>**2019-2023 Johnson Lab / Linksvayer Lab | Mentor, Researcher**

Research

- Postdoctoral position during NSF fellowship with Brian Johnson at UC Davis and Tim Linksvayer at Arizona State University.

Source: Johnson Lab <https://johnsonlab.ucdavis.edu/> | UC Davis <https://www.ucdavis.edu/>**2014-2019 Gordon Lab | Mentor, Researcher**

Research

- Graduate research, field work, wet lab, bioinformatics, and mentoring of high schoolers, undergraduates, graduate students, and non-academics.

Source: Gordon Lab <https://web.stanford.edu/~dmgordon/>**2010-2014 Kopp Lab | Researcher**

Research

- Undergraduate research assistant with Prof. Artyom Kopp in Ecology and Evolution at UC Davis, studying non-coding DNA elements in the evolution of the Drosophila foreleg sex comb.

Source: UC Davis Biology <https://biology.ucdavis.edu/>**2014-2019 Pedicab People Movers | Operator**

Pedicab

- Operated a pedicab/rickshaw at events including music festivals in California and downtown areas in Davis and San Jose.

Source: verification <https://danielarifriedman.com/resume/verify.html>**2015-2016 Stanford University | Teacher, Organizer, Graduate Teaching Assistant**

Teaching

- Graduate Teaching Assistant for undergraduate and graduate courses: General Biology: Cell and Molecular Biology (2015) and Ethics of Evolutionary Biology (2016).

Source: Stanford Biology <https://biologyv.stanford.edu/>**2014 UC Davis | Teacher, Organizer, Course Co-developer**

Teaching

- Student teacher and developmental biology course co-developer with Prof. Susan Keen at UC Davis.

Source: UC Davis Biology <https://biology.ucdavis.edu/>

## Awards and Fellowships (12)

| Year | Award                                                                                                                          | Details                                         |
|------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 2020 | <b>NSF Postdoctoral Research Fellowship in Biology</b>                                                                         | Award DBI-2010290; NSF budget period 2020-2022. |
|      | Source: NSF DBI-2010290<br><a href="https://grantome.com/grant/NSF/DBI-2010290">https://grantome.com/grant/NSF/DBI-2010290</a> |                                                 |



| Year | Award                                                                                                                                                                                                                                                                                                                                                                                        | Details                                                                                       |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 2014 | <b>Stanford Graduate Fellow, Urbanek Family Fellowship</b><br>Source: Stanford Graduate Fellowship<br><a href="https://vpqe.stanford.edu/fellowships-funding/sqf">https://vpqe.stanford.edu/fellowships-funding/sqf</a>                                                                                                                                                                      | Stanford University, 2014-2017.                                                               |
| 2019 | <b>Stanford Neuroscience NeuroChoice Grant</b><br>Source: Stanford Neurosciences Institute<br><a href="https://neuroscience.stanford.edu/">https://neuroscience.stanford.edu/</a>                                                                                                                                                                                                            | Neurometabolism in ants, \$25,000.                                                            |
| 2018 | <b>Stanford Systems Biology Seed Grant</b><br>Source: Stanford Systems Biology<br><a href="https://sysbio.stanford.edu/">https://sysbio.stanford.edu/</a>                                                                                                                                                                                                                                    | Gene expression and behavior in ants, \$25,000.                                               |
| 2016 | <b>Theodore Roosevelt Memorial Award</b><br>Source: AMNH Theodore Roosevelt Grants <a href="https://www.amnh.org/research/richard-gilder-graduate-school/academics-and-research/fellowships-and-grants/theodore-roosevelt-memorial-grants">https://www.amnh.org/research/richard-gilder-graduate-school/academics-and-research/fellowships-and-grants/theodore-roosevelt-memorial-grants</a> | American Museum of Natural History, \$2,500.                                                  |
| 2016 | <b>Lewis and Clark Fund grant</b><br>Source: APS Lewis and Clark Fund <a href="https://www.amp-hilsoc.org/grants/lewis-and-clark-fund-exploration-and-field-research">https://www.amp-hilsoc.org/grants/lewis-and-clark-fund-exploration-and-field-research</a>                                                                                                                              | American Philosophical Society, \$4,000.                                                      |
| 2015 | <b>NSF GRFP Honorable Mention</b><br>Source: NSF GRFP <a href="https://www.nsfgrfp.org/">https://www.nsfgrfp.org/</a>                                                                                                                                                                                                                                                                        | Honorable Mention in 2015 and 2016.                                                           |
| 2014 | <b>UC Davis Genetics Department Citation</b><br>Source: UC Davis Genetics<br><a href="https://genetics.ucdavis.edu/">https://genetics.ucdavis.edu/</a>                                                                                                                                                                                                                                       | Highest Honors.                                                                               |
| 2014 | <b>UC Davis service awards</b><br>Source: UC Davis <a href="https://www.ucdavis.edu/">https://www.ucdavis.edu/</a>                                                                                                                                                                                                                                                                           | Campus Community Service Silver Award and Stanley and Jacqueline Schilling Award for Service. |
| 2013 | <b>Victoria Finnerty Award for Research in Drosophila</b><br>Source: UC Davis Genetics<br><a href="https://genetics.ucdavis.edu/">https://genetics.ucdavis.edu/</a>                                                                                                                                                                                                                          | UC Davis.                                                                                     |
| 2014 | <b>Phi Beta Kappa</b><br>Source: Phi Beta Kappa <a href="https://www.pbk.org/">https://www.pbk.org/</a>                                                                                                                                                                                                                                                                                      | Elected member, UC Davis Chapter.                                                             |
| 2011 | <b>Provost's Undergraduate Fellowship</b><br>Source: UC Davis Undergraduate Research Center<br><a href="https://urc.ucdavis.edu/puf">https://urc.ucdavis.edu/puf</a>                                                                                                                                                                                                                         | Drosophila research, \$2,500.                                                                 |



## Works and Publications (193)

| #   | Year | Type         | Title and Venue                                                                                                                              | Link                                                                                                                                                |
|-----|------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 001 | 2026 | Paper        | <b>A template/ approach to Reproducible Generative Research: Architecture and Ergonomics from Configuration through Publication   Zenodo</b> | <a href="https://zenodo.org/record/16903351">10.5281/zenodo.16903351</a>                                                                            |
| 002 | 2026 | Series       | <b>Active Inference Journal — 500+ videos on Active Inference with transcripts   YouTube</b>                                                 | <a href="https://video.activeinference.institute/">https://video.activeinference.institute/</a>                                                     |
| 003 | 2026 | Paper        | <b>Before Pragmatism Had a Name: Blake's "America A Prophecy" Anticipates American Anticipatory Epistemology   Zenodo</b>                    | <a href="https://zenodo.org/record/18807970">10.5281/zenodo.18807970</a>                                                                            |
| 004 | 2026 | Paper        | <b>The Golden Compass and the Lunar Flux: William Blake, Bimetallic Meta-stability, and the Architecture of Value   Zenodo</b>               | <a href="https://zenodo.org/record/19335195">10.5281/zenodo.19335195</a>                                                                            |
| 005 | 2026 | Paper        | <b>Cognitive Integrity Framework: Formal Foundations for Multiagent Security (Part 1: Theory)   Zenodo</b>                                   | <a href="https://zenodo.org/record/18364118">10.5281/zenodo.18364118</a>                                                                            |
| 006 | 2026 | Series       | <b>Personal YouTube — 200+ livestreams: drawings, Synergetics, paper discussions   YouTube</b>                                               | <a href="https://www.youtube.com/@danielarifriedman/playlists">https://www.youtube.com/@danielarifriedman/playlists</a>                             |
| 007 | 2026 | Paper        | <b>The Doors of Perception are the Threshold of Prediction: Active Inference and William Blake's Theory of Seeing   Zenodo</b>               | <a href="https://zenodo.org/record/18600040">10.5281/zenodo.18600040</a>                                                                            |
| 008 | 2025 | Presentation | <b>5th Applied Active Inference Symposium — Abstract Book   Presentation</b>                                                                 | <a href="https://zenodo.org/record/17555266">10.5281/zenodo.17555266</a>                                                                            |
| 009 | 2025 | Paper        | <b>Adaptive Basic Income (AuBI): Integrating AI, Decentralized Infrastructure, and Active Inference   Zenodo</b>                             | <a href="https://zenodo.org/record/17228945">10.5281/zenodo.17228945</a>                                                                            |
| 010 | 2025 | Paper        | <b>CEREBRUM: Case-Enabled Reasoning Engine with Bayesian Representations for Unified Modeling   Zenodo</b>                                   | <a href="https://zenodo.org/record/15170907">10.5281/zenodo.15170907</a>                                                                            |
| 011 | 2025 | Paper        | <b>Computational Complexity and Energetics of the Ant Stack   Zenodo</b>                                                                     | <a href="https://zenodo.org/record/17238736">10.5281/zenodo.17238736</a>                                                                            |
| 012 | 2025 | Paper        | <b>EvoJump: Stochastic Modeling of Evolutionary Ontogenetic Trajectories   Zenodo</b>                                                        | <a href="https://zenodo.org/record/17229924">10.5281/zenodo.17229924</a>                                                                            |
| 013 | 2025 | Paper        | <b>Increasing the Accessibility and Applicability of Active Inference   Zenodo</b>                                                           | <a href="https://zenodo.org/record/15061666">10.5281/zenodo.15061666</a>                                                                            |
| 014 | 2025 | Paper        | <b>MDKV: A Multitrack Markdown Container for Structured, Portable Documents   Zenodo</b>                                                     | <a href="https://zenodo.org/record/16790554">10.5281/zenodo.16790554</a>                                                                            |
| 015 | 2025 | Paper        | <b>Markdown Decision Process: A Framework for Probabilistic Document Analysis   Zenodo</b>                                                   | <a href="https://zenodo.org/record/17244386">10.5281/zenodo.17244386</a>                                                                            |
| 016 | 2025 | Paper        | <b>On Cognitive Art &amp; Science: Toward Wholeness From Both Sides   Zenodo</b>                                                             | <a href="https://zenodo.org/record/16740438">10.5281/zenodo.16740438</a>                                                                            |
| 017 | 2025 | Paper        | <b>On Time   Zenodo</b>                                                                                                                      | <a href="https://zenodo.org/record/15168381">10.5281/zenodo.15168381</a>                                                                            |
| 018 | 2025 | Paper        | <b>QuadMath: An Analytical Review of 4D and Quadray Coordinates   Zenodo</b>                                                                 | <a href="https://zenodo.org/record/16887799">10.5281/zenodo.16887799</a>                                                                            |
| 019 | 2025 | Paper        | <b>ResNei: Solution Design Document   Zenodo</b>                                                                                             | <a href="https://zenodo.org/record/15389682">10.5281/zenodo.15389682</a>                                                                            |
| 020 | 2025 | Paper        | <b>Symergetics: Symbolic Synergetics for Rational Arithmetic   Zenodo</b>                                                                    | <a href="https://zenodo.org/record/17114389">10.5281/zenodo.17114389</a>                                                                            |
| 021 | 2025 | Paper        | <b>Synthesis of Agent and Niche   Zenodo</b>                                                                                                 | <a href="https://zenodo.org/record/17235137">10.5281/zenodo.17235137</a>                                                                            |
| 022 | 2025 | Presentation | <b>Systems Processes, Active Inference, and Beyond   Presentation</b>                                                                        | <a href="https://zenodo.org/record/17138223">10.5281/zenodo.17138223</a>                                                                            |
| 023 | 2025 | Paper        | <b>Temporal Depth in a Coherent Self and in Depersonalization   Frontiers in Psychology</b>                                                  | <a href="https://doi.org/10.3389/fpsyg.2025.1585315">10.3389/fpsyg.2025.1585315</a>                                                                 |
| 024 | 2025 | Paper        | <b>The Active Inference Institute &amp; Active Inference Ecosystem (v3, 2025 snapshot)   Zenodo</b>                                          | <a href="https://zenodo.org/record/17982447">10.5281/zenodo.17982447</a>                                                                            |
| 025 | 2025 | Paper        | <b>The Ant Stack   Zenodo</b>                                                                                                                | <a href="https://zenodo.org/record/16782756">10.5281/zenodo.16782756</a>                                                                            |
| 026 | 2025 | Paper        | <b>The Discovery Engine: AI-Driven Synthesis and Navigation of Scientific Knowledge Landscapes   ArXiv</b>                                   | <a href="https://arxiv.org/abs/2505.17500">10.48550/arXiv.2505.17500</a>                                                                            |
| 027 | 2025 | Paper        | <b>Thoughtseeds: A Hierarchical and Agentic Framework for Investigating Thought Dynamics in Meditative States   Entropy</b>                  | <a href="https://arxiv.org/abs/27050459">10.3390/e27050459</a>                                                                                      |
| 028 | 2025 | Paper        | <b>Towards a Science of Consciousness and Social Complexity... For Ants   Book Chapter</b>                                                   | <a href="https://www.editorafi.org/ebook/c128-colonias-formigas-conscientes">https://www.editorafi.org/ebook/c128-colonias-formigas-conscientes</a> |
| 029 | 2024 | Paper        | <b>Aligning Active Inference Ontology to SUMO   Zenodo</b>                                                                                   | <a href="https://zenodo.org/record/11459322">10.5281/zenodo.11459322</a>                                                                            |



| #   | Year | Type         | Title and Venue                                                                                                                                 | Link                                                                                                                      |
|-----|------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 030 | 2024 | Presentation | <b>BioFirm Development at Applied Active Inference Symposium 2024   Presentation</b>                                                            | <a href="https://zenodo.org/record/14861595">10.5281/zenodo.14861595</a>                                                  |
| 031 | 2024 | Paper        | <b>Bridging gaps in image meme research: A multidisciplinary paradigm   JASIST</b>                                                              | <a href="https://arxiv.org/abs/2409.10002">10.1002/asi.24900</a>                                                          |
| 032 | 2024 | Paper        | <b>Chemical and transcriptomic diversity do not correlate with ascending levels of social complexity in Blattodea   Ecology &amp; Evolution</b> | <a href="https://e3.ce3.org/doi/10.1002/ece3.70063">10.1002/ece3.70063</a>                                                |
| 033 | 2024 | Paper        | <b>Comments on National Digital Twins R&amp;D Strategic Plan   Zenodo</b>                                                                       | <a href="https://zenodo.org/record/13273681">10.5281/zenodo.13273681</a>                                                  |
| 034 | 2024 | Paper        | <b>Enhancing Population-based Search with Active Inference   ArXiv</b>                                                                          | <a href="https://arxiv.org/abs/2408.48550">10.48550/arXiv.2408.09548</a>                                                  |
| 035 | 2024 | Paper        | <b>FarmWorks: Decentralized AI Agents for Personalized Solutions   Zenodo</b>                                                                   | <a href="https://zenodo.org/record/13754585">10.5281/zenodo.13754585</a>                                                  |
| 036 | 2024 | Paper        | <b>Federated inference and belief sharing   Neuroscience &amp; Biobehavioral Reviews</b>                                                        | <a href="https://neurosci.biomedcentral.com/articles/10.1016/j.neubiorev.2023.105500">10.1016/j.neubiorev.2023.105500</a> |
| 037 | 2024 | Paper        | <b>Four-fold Fields of Quantum Dreams   Zenodo</b>                                                                                              | <a href="https://zenodo.org/record/10798144">10.5281/zenodo.10798144</a>                                                  |
| 038 | 2024 | Presentation | <b>MathArt Stream #8: William Blake and Active Inference   Presentation</b>                                                                     | <a href="https://zenodo.org/record/13711301">10.5281/zenodo.13711301</a>                                                  |
| 039 | 2024 | Presentation | <b>Sensemaking Federation: Exploring the Frontiers of Digital Innovation   Presentation</b>                                                     | <a href="https://zenodo.org/record/14574046">10.5281/zenodo.14574046</a>                                                  |
| 040 | 2024 | Paper        | <b>Shared Protentions in Multi-Agent Active Inference   Entropy</b>                                                                             | <a href="https://entropy.mindworksgroup.com/doi/10.3390/e26040303">10.3390/e26040303</a>                                  |
| 041 | 2024 | Paper        | <b>The Active Inference Institute &amp; Active Inference Ecosystem (v2, 2024 snapshot)   Zenodo</b>                                             | <a href="https://zenodo.org/record/14108992">10.5281/zenodo.14108992</a>                                                  |
| 042 | 2024 | Paper        | <b>Way Finding in the Infinite Imaginarium   Zenodo</b>                                                                                         | <a href="https://zenodo.org/record/10601081">10.5281/zenodo.10601081</a>                                                  |
| 043 | 2024 | Paper        | <b>Why Paleolithic Rockstars were both enigmatic and sporadic   Physics of Life Reviews</b>                                                     | <a href="https://plrev.org/doi/10.1016/j.plrev.2024.04.010">10.1016/j.plrev.2024.04.010</a>                               |
| 044 | 2024 | Paper        | <b>Writing on Curio Cards for the "On NFT" book   Taschen</b>                                                                                   | <a href="https://www.worldcat.org/isbn/978-3-8365-9970-2">https://www.worldcat.org/isbn/978-3-8365-9970-2</a>             |
| 045 | 2023 | Paper        | <b>A guided tour through the spaces of particular "minds"   Physics of Life Reviews</b>                                                         | <a href="https://plrev.org/doi/10.1016/j.plrev.2023.11.001">10.1016/j.plrev.2023.11.001</a>                               |
| 046 | 2023 | Paper        | <b>A single-pheromone model accounts for empirical patterns of ant colony foraging   Cognitive Systems Research</b>                             | <a href="https://cogsys.org/doi/10.1016/j.cogsys.2023.02.005">10.1016/j.cogsys.2023.02.005</a>                            |
| 047 | 2023 | Paper        | <b>A snapshot and pipeline for tissue-specific gene expression meta-analysis in honey bees   Zenodo</b>                                         | <a href="https://zenodo.org/record/10400744">10.5281/zenodo.10400744</a>                                                  |
| 048 | 2023 | Paper        | <b>A variational synthesis of evolutionary and developmental dynamics   Entropy</b>                                                             | <a href="https://entropy.mindworksgroup.com/doi/10.3390/e25070964">10.3390/e25070964</a>                                  |
| 049 | 2023 | Paper        | <b>ATLAS: A Question Oriented Approach to Pattern Languages in Knowledge Management   Zenodo</b>                                                | <a href="https://zenodo.org/record/10296601">10.5281/zenodo.10296601</a>                                                  |
| 050 | 2023 | Paper        | <b>An Account of Active Inference Modeling   Zenodo</b>                                                                                         | <a href="https://zenodo.org/record/8415312">10.5281/zenodo.8415312</a>                                                    |
| 051 | 2023 | Paper        | <b>Cognitive Sovereignty &amp; Active Inference in the State of Exception   Zenodo</b>                                                          | <a href="https://zenodo.org/record/10038231">10.5281/zenodo.10038231</a>                                                  |
| 052 | 2023 | Paper        | <b>Comments on AI Accountability Policy to NTIA   NTIA</b>                                                                                      | <a href="https://zenodo.org/record/8025956">10.5281/zenodo.8025956</a>                                                    |
| 053 | 2023 | Paper        | <b>Distributed Science — The Scientific Process as Multi-Scale Active Inference   OSF</b>                                                       | <a href="https://osf.io/dnw5k">10.31219/osf.io/dnw5k</a>                                                                  |
| 054 | 2023 | Paper        | <b>Enhanced NSF Postdoctoral Reporting via Synthetic Intelligence Language Processing   Zenodo</b>                                              | <a href="https://zenodo.org/record/10160656">10.5281/zenodo.10160656</a>                                                  |
| 055 | 2023 | Paper        | <b>Experimental Entomology in the Age of Video   JoVE</b>                                                                                       | <a href="https://doi.org/10.3791/65002">10.3791/65002</a>                                                                 |
| 056 | 2023 | Paper        | <b>Generalized Notation Notation for Active Inference Models   Zenodo</b>                                                                       | <a href="https://zenodo.org/record/7803327">10.5281/zenodo.7803327</a>                                                    |
| 057 | 2023 | Paper        | <b>Generative Research Teams: Active Inference Compositions for Research and Meta-Science   Zenodo</b>                                          | <a href="https://zenodo.org/record/8164666">10.5281/zenodo.8164666</a>                                                    |
| 058 | 2023 | Presentation | <b>Of Ants &amp; Aging   Presentation</b>                                                                                                       | <a href="https://zenodo.org/record/7855581">10.5281/zenodo.7855581</a>                                                    |
| 059 | 2023 | Paper        | <b>Open Access science needs Open Science Sensemaking (OSSm)   MetaArXiv</b>                                                                    | <a href="https://osf.io/9nb3u">10.31222/osf.io/9nb3u</a>                                                                  |
| 060 | 2023 | Presentation | <b>Postdoc review (2020–2023)   Presentation</b>                                                                                                | <a href="https://zenodo.org/record/8377987">10.5281/zenodo.8377987</a>                                                    |
| 061 | 2023 | Paper        | <b>The Active Inference Institute and Active Inference Ecosystem (v1)   Zenodo</b>                                                              | <a href="https://zenodo.org/record/8266280">10.5281/zenodo.8266280</a>                                                    |



| #   | Year | Type     | Title and Venue                                                                                                                                   | Link                                                                                                                                    |
|-----|------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 062 | 2023 | Paper    | <b>The P3IF: Properties, Processes, and Perspectives Inter-Framework   Zenodo</b>                                                                 | <a href="https://zenodo.org/record/10034511">10.5281/zenodo.10034511</a>                                                                |
| 063 | 2023 | Paper    | <b>To comment or not to comment   Physics of Life Reviews</b>                                                                                     | <a href="https://www.ploprev.org/article/pii/plprev.2023.06.002">10.1016/j.plprev.2023.06.002</a>                                       |
| 064 | 2023 | Course   | <b>Transcript of Chris Fields, "Physics as Information Processing" course   Zenodo</b>                                                            | <a href="https://zenodo.org/record/10447642">10.5281/zenodo.10447642</a>                                                                |
| 065 | 2023 | Paper    | <b>William Blake &amp; Buckminster Fuller: Lives in Juxtaposition   Zenodo</b>                                                                    | <a href="https://zenodo.org/record/7514367">10.5281/zenodo.7514367</a>                                                                  |
| 066 | 2022 | Paper    | <b>An Active Inference Ontology for Decentralized Science   Zenodo</b>                                                                            | <a href="https://zenodo.org/record/6320574">10.5281/zenodo.6320574</a>                                                                  |
| 067 | 2022 | Paper    | <b>Catechism for Towards Active Diffusion   Zenodo</b>                                                                                            | <a href="https://zenodo.org/record/7443847">10.5281/zenodo.7443847</a>                                                                  |
| 068 | 2022 | Paper    | <b>From Users to (Sense)Makers: Stigmergic Social Annotation   Hypertext '22</b>                                                                  | <a href="https://arxiv.org/abs/2205.06345">10.48550/arXiv.2205.06345</a>                                                                |
| 069 | 2022 | Paper    | <b>On free will or the lack thereof (interview with Robert Sapolsky)   ALIUS Bulletin</b>                                                         | <a href="https://zenodo.org/record/7394900">10.5281/zenodo.7394900</a>                                                                  |
| 070 | 2022 | Paper    | <b>Predictive Processing Interpretation of the Mirror Test   Zenodo</b>                                                                           | <a href="https://zenodo.org/record/7377255">10.5281/zenodo.7377255</a>                                                                  |
| 071 | 2022 | Book     | <b>Structuring the Information Commons: Open Standards and Cognitive Security   COGSEC.org</b>                                                    | <a href="https://cogsec.org">https://cogsec.org</a>                                                                                     |
| 072 | 2022 | Paper    | <b>Systems Modeling and Cognitive Audits for Hypercert Ecosystems   Zenodo</b>                                                                    | <a href="https://zenodo.org/record/7626768">10.5281/zenodo.7626768</a>                                                                  |
| 073 | 2022 | Paper    | <b>The Free Energy Principle &amp; Active Inference: a Systematic Literature Analysis   Zenodo</b>                                                | <a href="https://zenodo.org/record/7449367">10.5281/zenodo.7449367</a>                                                                  |
| 074 | 2022 | Paper    | <b>TrustFinder: Recommendations for Community-Based Trust Systems   Zenodo</b>                                                                    | <a href="https://zenodo.org/record/7093836">10.5281/zenodo.7093836</a>                                                                  |
| 075 | 2021 | Paper    | <b>Active Inferants: An Active Inference Framework for Ant Colony Behavior   Frontiers in Behavioral Neuroscience</b>                             | <a href="https://www.frontiersin.org/article/10.3389/fnbeh.2021.647732">10.3389/fnbeh.2021.647732</a>                                   |
| 076 | 2021 | Paper    | <b>Active Inference in Modeling Conflict   Zenodo</b>                                                                                             | <a href="https://zenodo.org/record/5750934">10.5281/zenodo.5750934</a>                                                                  |
| 077 | 2021 | Paper    | <b>Digital Rhetorical Ecosystem Analysis: Sensemaking of Digital Memetic Discourse   Zenodo</b>                                                   | <a href="https://zenodo.org/record/5573946">10.5281/zenodo.5573946</a>                                                                  |
| 078 | 2021 | Playbook | <b>Innovator's Digital Playbook   Web</b>                                                                                                         | <a href="https://innovatorsdigitalplaybook.com/">https://innovatorsdigitalplaybook.com/</a>                                             |
| 079 | 2021 | Book     | <b>Narrative Information Ecosystems: Conflict and Trust on the Endless Frontier   COGSEC.org</b>                                                  | <a href="https://cogsec.org">https://cogsec.org</a>                                                                                     |
| 080 | 2021 | Paper    | <b>Narrative Information Management   Zenodo</b>                                                                                                  | <a href="https://zenodo.org/record/5565577">10.5281/zenodo.5565577</a>                                                                  |
| 081 | 2021 | Playbook | <b>The Facilitator's Catechism Playbook   Zenodo</b>                                                                                              | <a href="https://zenodo.org/record/4579413">10.5281/zenodo.4579413</a>                                                                  |
| 082 | 2020 | Paper    | <b>Active Inference &amp; Behavior Engineering for Teams   Zenodo</b>                                                                             | <a href="https://zenodo.org/record/4021162">10.5281/zenodo.4021162</a>                                                                  |
| 083 | 2020 | Course   | <b>Communication for Remote Teams   Udemy</b>                                                                                                     | <a href="https://www.udemy.com/course/communication-for-remote-teams/">https://www.udemy.com/course/communication-for-remote-teams/</a> |
| 084 | 2020 | Paper    | <b>Distributed physiology and the molecular basis of social life in eusocial insects   Hormones &amp; Behavior</b>                                | <a href="https://www.sciencedirect.com/science/article/pii/S0016718920300001">10.1016/j.yhbeh.2020.104757</a>                           |
| 085 | 2020 | Paper    | <b>Emergent Teams for Complex Threats   Zenodo</b>                                                                                                | <a href="https://zenodo.org/record/3986084">10.5281/zenodo.3986084</a>                                                                  |
| 086 | 2020 | Paper    | <b>Gene expression variation in the brains of harvester ant foragers is associated with collective behavior   Communications Biology</b>          | <a href="https://doi.org/10.1038/s42003-020-0813-8">10.1038/s42003-020-0813-8</a>                                                       |
| 087 | 2020 | Course   | <b>How to use Keybase for remote teams   Udemy</b>                                                                                                | <a href="https://www.udemy.com/course/keybase-for-remote-teams/">https://www.udemy.com/course/keybase-for-remote-teams/</a>             |
| 088 | 2020 | Paper    | <b>Measurement of natural variation of neurotransmitter tissue content in red harvester ant brains   Analytical &amp; Bioanalytical Chemistry</b> | <a href="https://doi.org/10.1007/s00216-019-02355-3">10.1007/s00216-019-02355-3</a>                                                     |
| 089 | 2020 | Book     | <b>The Great Preset: Remote Teams and Operational Art   COGSEC.org</b>                                                                            | <a href="https://cogsec.org">https://cogsec.org</a>                                                                                     |
| 090 | 2020 | Paper    | <b>The Innovator's Catechism   Zenodo</b>                                                                                                         | <a href="https://zenodo.org/record/4383229">10.5281/zenodo.4383229</a>                                                                  |
| 091 | 2020 | Paper    | <b>The neuroscience of decision making (interview with Timothy Hanks)   ALIUS Bulletin</b>                                                        | <a href="https://zenodo.org/record/347008/p1/1">10.34700/8pg4-0h12</a>                                                                  |
| 092 | 2019 | Paper    | <b>Dennett Explained (interview with Daniel Dennett)   ALIUS Bulletin</b>                                                                         | <a href="https://zenodo.org/record/347007/p1/1">10.34700/7qkw-zh08</a>                                                                  |
| 093 | 2019 | Paper    | <b>PhD: Behavioral, Physiological, and Transcriptomic Variation Among Colonies of Pogonomyrmex barbatus   Stanford University</b>                 | <a href="http://purl.stanford.edu/pb813wm1484">http://purl.stanford.edu/pb813wm1484</a>                                                 |
| 094 | 2019 | Paper    | <b>The ant colony as a test for scientific theories of consciousness   Synthese</b>                                                               | <a href="https://doi.org/10.1007/s11229-019-02130-y">10.1007/s11229-019-02130-y</a>                                                     |





| #   | Year | Type         | Title and Venue                                                                                                                                                                       | Link                                                                                                  |
|-----|------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 095 | 2019 | Paper        | <b>The physiology of forager hydration and variation among harvester ant colonies   Scientific Reports</b>                                                                            | <a href="https://doi.org/10.1038/s41598-019-41586-3">10.1038/s41598-019-41586-3</a>                   |
| 096 | 2018 | Paper        | <b>Foraging behavior and locomotion of the invasive Argentine ant from winter aggregations   PLoS One</b>                                                                             | <a href="https://doi.org/10.1371/journal.pone.0202117">10.1371/journal.pone.0202117</a>               |
| 097 | 2018 | Presentation | <b>MVEE: A Framework for Evolutionary Studies   Presentation</b>                                                                                                                      | <a href="https://doi.org/10.5281/zenodo.13999298">10.5281/zenodo.13999298</a>                         |
| 098 | 2018 | Paper        | <b>Of woodlice and men: A Bayesian account of cognition, life and consciousness (with Karl Friston)   ALIUS Bulletin</b>                                                              | <a href="https://doi.org/10.34700/h460-nz89">10.34700/h460-nz89</a>                                   |
| 099 | 2018 | Paper        | <b>Partner Pen Play in Parallel (PPPiP): A New Paradigm for Relationship Improvement   Arts</b>                                                                                       | <a href="https://doi.org/10.3390/arts7030039">10.3390/arts7030039</a>                                 |
| 100 | 2018 | Paper        | <b>The Role of Dopamine in Collective Regulation of Foraging in Harvester Ants   iScience</b>                                                                                         | <a href="https://doi.org/10.1016/j.isci.2018.09.001">10.1016/j.isci.2018.09.001</a>                   |
| 101 | 2017 | Paper        | <b>The MutAnts are here   Cell</b>                                                                                                                                                    | <a href="https://doi.org/10.1016/j.cell.2017.07.046">10.1016/j.cell.2017.07.046</a>                   |
| 102 | 2017 | Paper        | <b>Two lineages that need each other   Molecular Ecology</b>                                                                                                                          | <a href="https://doi.org/10.1111/mec.13964">10.1111/mec.13964</a>                                     |
| 103 | 2016 | Paper        | <b>Ant Genetics: Reproductive Physiology, Worker Morphology, and Behavior   Annual Review of Neuroscience</b>                                                                         | <a href="https://doi.org/10.1146/annurev-neuro-070815-013927">10.1146/annurev-neuro-070815-013927</a> |
| 104 | 2016 | Paper        | <b>Context-dependent expression of the foraging gene in field colonies of ants   Proceedings of the Royal Society B</b>                                                               | <a href="https://doi.org/10.1098/rspb.2016.0841">10.1098/rspb.2016.0841</a>                           |
| 105 | 2016 | Paper        | <b>Influence of Nuclear Structure on formation of radiation-induced lethal lesions   Int. J. Radiation Biology</b>                                                                    | <a href="https://doi.org/10.3109/09553002.2016.1144941">10.3109/09553002.2016.1144941</a>             |
| 106 | 2015 | Paper        | <b>Commentary: Portuguese crypto-Jews: the genetic heritage of a complex history   Frontiers in Genetics</b>                                                                          | <a href="https://doi.org/10.3389/fgene.2015.00261">10.3389/fgene.2015.00261</a>                       |
| 107 | 2015 | Paper        | <b>Could Ehrlichial Infection Cause Some Changes Associated with Leukemia?   Medical Hypotheses</b>                                                                                   | <a href="https://doi.org/10.1016/j.mehy.2015.09.015">10.1016/j.mehy.2015.09.015</a>                   |
| 108 | 2015 | Paper        | <b>Large Scale Coding Sequence Change Underlies the Evolution of Post-developmental Novelty in Honey Bees   Molecular Biology &amp; Evolution</b>                                     | <a href="https://doi.org/10.1093/molbev/msu292">10.1093/molbev/msu292</a>                             |
| 109 | 2026 | Paper        | <b>Ento-Linguistics: Language, Ambiguity, and Scientific Communication in Entomology   Zenodo</b>                                                                                     | <a href="https://doi.org/10.5281/zenodo.19574117">10.5281/zenodo.19574117</a>                         |
| 110 | 2026 | Paper        | <b>A Living Meta-Analysis Architecture for Active Inference: Assertion Extraction, Nanopublications, and Hypothesis Scoring   Active Inference Journal</b>                            | <a href="https://doi.org/10.5281/zenodo.19461933">10.5281/zenodo.19461933</a>                         |
| 111 | 2026 | Paper        | <b>Dynamic Attentional Agents in Focused Attention Meditation: Hierarchical Computational Modeling of Expert-Novice Differences   CSCIS vol 2857, Springer</b>                        | <a href="https://doi.org/10.1007/978-3-032-16955-6_11">10.1007/978-3-032-16955-6_11</a>               |
| 112 | 2026 | Paper        | <b>Compositional Approaches to Linguistic Case for Cognitive Modeling   Active Inference Journal</b>                                                                                  | <a href="https://doi.org/10.5281/zenodo.19695259">10.5281/zenodo.19695259</a>                         |
| 113 | 2026 | Paper        | <b>Towards Lean 4 Formalization of the Free Energy Principle: AI-Driven Theorem Sketching and Verification for Active Inference and Bayesian Mechanics   Active Inference Journal</b> | <a href="https://doi.org/10.5281/zenodo.19699233">10.5281/zenodo.19699233</a>                         |
| 114 | 2020 | Paper        | <b>The Facilitator's Catechism   Zenodo</b>                                                                                                                                           | <a href="https://doi.org/10.5281/zenodo.4062540">10.5281/zenodo.4062540</a>                           |
| 115 | 2026 | Paper        | <b>The Architecture of False Gods: William Blake, Professor Jiang, and the Active Inference Corrective to Single Vision   Zenodo</b>                                                  | <a href="https://doi.org/10.5281/zenodo.20144983">10.5281/zenodo.20144983</a>                         |
| 116 | 2026 | Paper        | <b>Crescent City in Living Waves: Space, Time, People, and Minds on the Southern Cascadian Coast   Zenodo</b>                                                                         | <a href="https://doi.org/10.5281/zenodo.20286170">10.5281/zenodo.20286170</a>                         |
| 117 | 2026 | Book         | <b>Introduction to Biology: A Generative Approach   Zenodo</b>                                                                                                                        | <a href="https://doi.org/10.5281/zenodo.20286477">10.5281/zenodo.20286477</a>                         |
| 119 | 2026 | Paper        | <b>Policy Entanglement in Active Inference   Zenodo</b>                                                                                                                               | <a href="https://doi.org/10.5281/zenodo.20418904">10.5281/zenodo.20418904</a>                         |
| 120 | 2026 | Paper        | <b>Bounded AutoResearch for a Tiny Reproducible Machine-Learning Task   Zenodo</b>                                                                                                    | <a href="https://doi.org/10.5281/zenodo.20417016">10.5281/zenodo.20417016</a>                         |
| 121 | 2026 | Paper        | <b>Convergence Analysis of Gradient Descent Optimization   Zenodo</b>                                                                                                                 | <a href="https://doi.org/10.5281/zenodo.20417136">10.5281/zenodo.20417136</a>                         |
| 122 | 2026 | Paper        | <b>Editorial Quality at Scale: A Reproducible Prose-Review Pipeline   Zenodo</b>                                                                                                      | <a href="https://doi.org/10.5281/zenodo.20417104">10.5281/zenodo.20417104</a>                         |
| 123 | 2026 | Paper        | <b>A template/ approach to Reproducible Generative Research   Zenodo</b>                                                                                                              | <a href="https://doi.org/10.5281/zenodo.20419007">10.5281/zenodo.20419007</a>                         |
| 124 | 2026 | Paper        | <b>Active Inference Multi-Track Exemplar   Zenodo</b>                                                                                                                                 | <a href="https://doi.org/10.5281/zenodo.20417021">10.5281/zenodo.20417021</a>                         |



| #   | Year | Type         | Title and Venue                                                                                                                                                    | Link                                                                                                                            |
|-----|------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 125 | 2026 | Paper        | <b>BeeStack: An Evidence-Typed Scaffold for Whole-Colony Honeybee Simulation   Zenodo</b>                                                                          | <a href="https://zenodo.org/record/20420556">10.5281/zenodo.20420556</a>                                                        |
| 126 | 2026 | Paper        | <b>When do bugs see (infra)red?   Zenodo</b>                                                                                                                       | <a href="https://zenodo.org/record/20450880">10.5281/zenodo.20450880</a>                                                        |
| 127 | 2026 | Paper        | <b>Self-Improvement Agent Harness: A Deterministic SIA Exemplar   Zenodo</b>                                                                                       | <a href="https://zenodo.org/record/20453879">10.5281/zenodo.20453879</a>                                                        |
| 128 | 2025 | Presentation | <b>Biofirm Development with First Principles First and the Active Inference Institute at the Applied Active Inference Symposium 2024   Zenodo</b>                  | <a href="https://zenodo.org/record/14861596">10.5281/zenodo.14861596</a>                                                        |
| 129 | 2025 | Paper        | <b>Con-cat-enate: emulation of Cat Hippocampus   Zenodo</b>                                                                                                        | <a href="https://zenodo.org/record/13626536">10.5281/zenodo.13626536</a>                                                        |
| 130 | 2025 | Paper        | <b>Graphspeak: of language and handshake   Zenodo</b>                                                                                                              | <a href="https://zenodo.org/record/14737156">10.5281/zenodo.14737156</a>                                                        |
| 131 | 2025 | Paper        | <b>BevCyc - posit on primitive drivers of creatures   Zenodo</b>                                                                                                   | <a href="https://zenodo.org/record/14737075">10.5281/zenodo.14737075</a>                                                        |
| 132 | 2025 | Paper        | <b>Beacons - pre emulation of social cortex   Zenodo</b>                                                                                                           | <a href="https://zenodo.org/record/14737059">10.5281/zenodo.14737059</a>                                                        |
| 133 | 2025 | Paper        | <b>Con-cat-enate: pilot overview   Zenodo</b>                                                                                                                      | <a href="https://zenodo.org/record/14737042">10.5281/zenodo.14737042</a>                                                        |
| 134 | 2024 | Paper        | <b>Push and Pull: A priming sequence   Zenodo</b>                                                                                                                  | <a href="https://zenodo.org/record/10659374">10.5281/zenodo.10659374</a>                                                        |
| 135 | 2023 | Paper        | <b>A Natural AI Based on The Science of Computational Physics, Biology and Neuroscience: Policy and Societal Significance   Zenodo</b>                             | <a href="https://zenodo.org/record/10360056">10.5281/zenodo.10360056</a>                                                        |
| 136 | 2023 | Paper        | <b>ATLAS: A Question Oriented Approach to the Use of Pattern Languages in Knowledge Management   Zenodo</b>                                                        | <a href="https://zenodo.org/record/10362561">10.5281/zenodo.10362561</a>                                                        |
| 137 | 2023 | Paper        | <b>Transcript of Active Inference GuestStream 049.1: "Clickbait, consciousness science, and responsible journalism"   Zenodo</b>                                   | <a href="https://zenodo.org/record/8228934">10.5281/zenodo.8228934</a>                                                          |
| 138 | 2023 | Paper        | <b>Modern Nostr Index Card-based Knowledge Engineering   Zenodo</b>                                                                                                | <a href="https://zenodo.org/record/8118155">10.5281/zenodo.8118155</a>                                                          |
| 139 | 2023 | Paper        | <b>Slides for Iris   Zenodo</b>                                                                                                                                    | <a href="https://zenodo.org/record/7838652">10.5281/zenodo.7838652</a>                                                          |
| 140 | 2022 | Paper        | <b>Canonical neural networks perform active inference   Zenodo</b>                                                                                                 | <a href="https://zenodo.org/record/7400525">10.5281/zenodo.7400525</a>                                                          |
| 141 | 2022 | Paper        | <b>Interoception as modeling, allostasis as control   Zenodo</b>                                                                                                   | <a href="https://zenodo.org/record/7400708">10.5281/zenodo.7400708</a>                                                          |
| 142 | 2022 | Paper        | <b>Transcript of: Mark Solms, "Consciousness as Precision Optimization: Some Physiological and Philosophical Considerations", ActInf GuestStream #016   Zenodo</b> | <a href="https://zenodo.org/record/7259006">10.5281/zenodo.7259006</a>                                                          |
| 143 | 2022 | Paper        | <b>A Worked Example of the Bayesian Mechanics of Classical Objects   Zenodo</b>                                                                                    | <a href="https://zenodo.org/record/7400785">10.5281/zenodo.7400785</a>                                                          |
| 144 | 2022 | Paper        | <b>Transcript of discussions on: "Communication as Socially Extended Active Inference: An Ecological Approach to Communicative Behavior"   Zenodo</b>              | <a href="https://zenodo.org/record/7401874">10.5281/zenodo.7401874</a>                                                          |
| 145 | 2022 | Paper        | <b>Tracking Public Sensemaking through Rhetorical Annotation of Image Memes   Zenodo</b>                                                                           | <a href="https://zenodo.org/record/6904426">10.5281/zenodo.6904426</a>                                                          |
| 146 | 2021 | Paper        | <b>Transcript of: Karl Friston, 1st Applied Active Inference Symposium, Active Inference Lab, June 21, 2021   Zenodo</b>                                           | <a href="https://zenodo.org/record/5797040">10.5281/zenodo.5797040</a>                                                          |
| 147 | 2021 | Paper        | <b>Collaborative Writing for Catechism-Based Teams   Zenodo</b>                                                                                                    | <a href="https://zenodo.org/record/4624298">10.5281/zenodo.4624298</a>                                                          |
| 148 | 2020 | Paper        | <b>Reimagining Maps   Zenodo</b>                                                                                                                                   | <a href="https://zenodo.org/record/4170025">10.5281/zenodo.4170025</a>                                                          |
| 149 | 2020 | Paper        | <b>Infinite Games for Infinite Teams   Zenodo</b>                                                                                                                  | <a href="https://zenodo.org/record/12601674">10.5281/zenodo.12601674</a>                                                        |
| 150 | 2026 | Paper        | <b>The Music Never Stopped: A Grateful Data Compendium with a Category-Theoretic Interpretation   Zenodo</b>                                                       | <a href="https://zenodo.org/record/20482025">10.5281/zenodo.20482025</a>                                                        |
| 151 | 2026 | Paper        | <b>A Deterministic Testbed for Self-Organizing Agent-Team Coordination   Zenodo</b>                                                                                | <a href="https://zenodo.org/record/20533669">10.5281/zenodo.20533669</a>                                                        |
| 152 | 2026 | Paper        | <b>Recovering LLM-Persona Accuracies from Unlabeled Votes   Zenodo</b>                                                                                             | <a href="https://zenodo.org/record/20498699">10.5281/zenodo.20498699</a>                                                        |
| 153 | 2026 | Paper        | <b>The Triplicate: A Data-Driven Large-Format Newspaper Layout Engine   Zenodo</b>                                                                                 | <a href="https://zenodo.org/record/20533675">10.5281/zenodo.20533675</a>                                                        |
| 154 | 2026 | Book         | <b>The Template Textbook   Zenodo</b>                                                                                                                              | <a href="https://zenodo.org/record/20533125">10.5281/zenodo.20533125</a>                                                        |
| 155 | 2016 | Paper        | <b>Lessons from the colony   Science</b>                                                                                                                           | <a href="https://www.science.org/doi/10.1126/science.352.6288.886">https://www.science.org/doi/10.1126/science.352.6288.886</a> |
| 156 | 2016 | Paper        | <b>Full speed ahead to the City on the Hill   Science</b>                                                                                                          | <a href="https://www.science.org/doi/10.1126/science.352.6288.886">10.1126/science.352.6288.886</a>                             |
| 157 | 2021 | Paper        | <b>Disinforge   Science</b>                                                                                                                                        | <a href="https://www.science.org/doi/10.1126/science.371.6524.22">https://www.science.org/doi/10.1126/science.371.6524.22</a>   |
| 158 | 2021 | Paper        | <b>Defining events: 2020 in hindsight   Science</b>                                                                                                                | <a href="https://www.science.org/doi/10.1126/science.371.6524.22">10.1126/science.371.6524.22</a>                               |



| #   | Year | Type  | Title and Venue                                                                                                                                 | Link                                                                                    |
|-----|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 159 | 2022 | Paper | <b>Active Blockference: cadCAD with Active Inference for Cognitive Systems Modeling   International Workshop on Active Inference</b>            | <a href="https://doi.org/10.1007/978-3-031-28719-0_10">10.1007/978-3-031-28719-0_10</a> |
| 160 | 2024 | Paper | <b>From Neuronal Packets to Thoughtseeds: A Hierarchical Model of Embodied Cognition in the Global Workspace   arXiv</b>                        | <a href="https://arxiv.org/abs/10.48550/arXiv.2408.15982">10.48550/arXiv.2408.15982</a> |
| 161 | 2021 | Paper | <b>Knowledge Management Archipelago   Zenodo</b>                                                                                                | <a href="https://zenodo.org/record/5034808">10.5281/zenodo.5034808</a>                  |
| 162 | 2024 | Paper | <b>Comments Submitted by BlockScience, UW APL IRSIRI, and the Cognitive Security and Education Forum   Zenodo</b>                               | <a href="https://zenodo.org/record/13273682">10.5281/zenodo.13273682</a>                |
| 163 | 2017 | Paper | <b>Osteopathy and the Variable Variability of Health   The Cranial Letter 70(4)</b>                                                             |                                                                                         |
| 164 | 2016 | Paper | <b>Cells, Mechanobiology, and Osteopathy   The Cranial Letter 69(1)</b>                                                                         |                                                                                         |
| 165 | 2026 | Paper | <b>iTrace: verification-first webcam eye-movement analysis   Zenodo</b>                                                                         | <a href="https://zenodo.org/record/20614908">10.5281/zenodo.20614908</a>                |
| 166 | 2026 | Paper | <b>DemoCreate: Declarative Audio-Visual Demo Generation for Software   Zenodo</b>                                                               | <a href="https://zenodo.org/record/20693216">10.5281/zenodo.20693216</a>                |
| 167 | 2026 | Paper | <b>ENTO: an ENcrypted, Typed, Omnitrack container format for multimodal research data   Zenodo</b>                                              | <a href="https://zenodo.org/record/20396328">10.5281/zenodo.20396328</a>                |
| 168 | 2026 | Paper | <b>GeneralizedNotationNotation (GNN)   Zenodo</b>                                                                                               | <a href="https://zenodo.org/record/7803313">10.5281/zenodo.7803313</a>                  |
| 169 | 2026 | Paper | <b>COGANT: Deterministic Codebase-to-GNN Translation   Zenodo</b>                                                                               | <a href="https://zenodo.org/record/20705350">10.5281/zenodo.20705350</a>                |
| 170 | 2026 | Paper | <b>AGEINT: Agentic Intelligence   Zenodo</b>                                                                                                    | <a href="https://zenodo.org/record/20732274">10.5281/zenodo.20732274</a>                |
| 171 | 2026 | Paper | <b>On-Policy Distillation as Active Inference in Finite Variational Models   Zenodo</b>                                                         | <a href="https://zenodo.org/record/20747834">10.5281/zenodo.20747834</a>                |
| 172 | 2026 | Paper | <b>Template Madlib: Deterministic Token Injection for Conditional IMRAD Manuscripts   Zenodo</b>                                                | <a href="https://zenodo.org/record/20786638">10.5281/zenodo.20786638</a>                |
| 173 | 2026 | Paper | <b>California Public Records: A Technical and Legal Reference for the Post-AB 473 Era   Zenodo</b>                                              | <a href="https://zenodo.org/record/20789899">10.5281/zenodo.20789899</a>                |
| 174 | 2026 | Paper | <b>CogSecSkills: Multiharness Cognitive Security Skill Library   Zenodo</b>                                                                     | <a href="https://zenodo.org/record/20804585">10.5281/zenodo.20804585</a>                |
| 175 | 2026 | Paper | <b>Realizing Emptiness: Operational Surrogates for No-Self-Evidence, QRF Opacification, and Bayesian Model Reduction   Zenodo</b>               | <a href="https://zenodo.org/record/20834846">10.5281/zenodo.20834846</a>                |
| 176 | 2026 | Paper | <b>A Living Meta-Analysis of the Modafinil Literature   Zenodo</b>                                                                              | <a href="https://zenodo.org/record/20931964">10.5281/zenodo.20931964</a>                |
| 177 | 2026 | Paper | <b>Refinement of Gold: A Metallurgical Analogy for Scientific Manuscript Composition   Zenodo</b>                                               | <a href="https://zenodo.org/record/20931955">10.5281/zenodo.20931955</a>                |
| 178 | 2026 | Paper | <b>AlphaCOGANT: Recursive Corporate Self-Improvement as Active Inference   Zenodo</b>                                                           | <a href="https://zenodo.org/record/20976824">10.5281/zenodo.20976824</a>                |
| 179 | 2026 | Paper | <b>Mapping William Blake's Works: Evidence ledgers, source provenance, text-image diagnostics, and rights-bounded release controls   Zenodo</b> | <a href="https://zenodo.org/record/21047573">10.5281/zenodo.21047573</a>                |
| 180 | 2026 | Paper | <b>Sortition Upstream of NTQR   Zenodo</b>                                                                                                      | <a href="https://zenodo.org/record/21083779">10.5281/zenodo.21083779</a>                |
| 181 | 2026 | Paper | <b>Exploratory Data Analysis: A Reproducible Notebook Template   Zenodo</b>                                                                     | <a href="https://zenodo.org/record/21086292">10.5281/zenodo.21086292</a>                |
| 182 | 2026 | Paper | <b>A Domain Language for Specifying Controlled Methods   Zenodo</b>                                                                             | <a href="https://zenodo.org/record/21086548">10.5281/zenodo.21086548</a>                |
| 183 | 2026 | Paper | <b>Entomological Law: A Field Map of Insects as Evidence, Threat, Property, Product, Patient, and Weapon   Zenodo</b>                           | <a href="https://zenodo.org/record/21137276">10.5281/zenodo.21137276</a>                |
| 184 | 2026 | Paper | <b>The Shape Between: A Full-Page Illustrated Storybook Template   Zenodo</b>                                                                   | <a href="https://zenodo.org/record/21176000">10.5281/zenodo.21176000</a>                |
| 185 | 2026 | Paper | <b>Autopoietic Project Generation   Zenodo</b>                                                                                                  | <a href="https://zenodo.org/record/21227869">10.5281/zenodo.21227869</a>                |
| 186 | 2026 | Paper | <b>template_pitch_deck: Reproducible, Validated Pitch-Deck Generation   Zenodo</b>                                                              | <a href="https://zenodo.org/record/21281509">10.5281/zenodo.21281509</a>                |
| 187 | 2026 | Paper | <b>Reproducible Literature Synthesis with infrastructure/search and infrastructure/reference   Zenodo</b>                                       | <a href="https://zenodo.org/record/21298894">10.5281/zenodo.21298894</a>                |
| 188 | 2026 | Paper | <b>Registered Report Template: Preregistration, Deviations, and Claim Boundaries   Zenodo</b>                                                   | <a href="https://zenodo.org/record/21298892">10.5281/zenodo.21298892</a>                |
| 189 | 2026 | Paper | <b>Redacted Report Template: Disclosure Control and Release Audit   Zenodo</b>                                                                  | <a href="https://zenodo.org/record/21298890">10.5281/zenodo.21298890</a>                |



| #   | Year | Type  | Title and Venue                                                                                                                              | Link                                                                     |
|-----|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 190 | 2026 | Paper | <b>Pools, Rules, and Tools: A Template-Integrated Resource Architecture   Zenodo</b>                                                         | <a href="https://zenodo.org/record/2129888">10.5281/zenodo.2129888</a>   |
| 191 | 2026 | Paper | <b>Illegal States, Mostly Unrepresentable   Zenodo</b>                                                                                       | <a href="https://zenodo.org/record/2129885">10.5281/zenodo.2129885</a>   |
| 192 | 2026 | Paper | <b>Data Descriptor Template: Schema, Provenance, and Release Readiness   Zenodo</b>                                                          | <a href="https://zenodo.org/record/2129883">10.5281/zenodo.2129883</a>   |
| 193 | 2026 | Paper | <b>A template/ approach to Reproducible Generative Research: Architecture and Ergonomics from Configuration through Publication   Zenodo</b> | <a href="https://zenodo.org/record/19139089">10.5281/zenodo.19139089</a> |
| 194 | 2026 | Paper | <b>Prior Cognitive Art   Zenodo</b>                                                                                                          | <a href="https://zenodo.org/record/21316510">10.5281/zenodo.21316510</a> |



## Software (95)

| Repository                                                                                                                                                                                  | Stack            | Description and Source Links                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>docxology/template</b><br><a href="https://github.com/docxology/template">https://github.com/docxology/template</a>                                                                      | Python           | Production-grade scaffold for reproducible computational research — 10-stage DAG pipelines, ≥90% test coverage, cryptographic provenance, multi-project workspace, and agent-ready documentation · Zenodo software v3.3.0 · Zenodo<br><br>paper: papers/2026_AutopoieticProjectGeneration/   zenodo:<br><a href="https://doi.org/10.5281/zenodo.20584820">https://doi.org/10.5281/zenodo.20584820</a> |
| <b>docxology/QuadCraft</b><br><a href="https://github.com/docxology/QuadCraft">https://github.com/docxology/QuadCraft</a>                                                                   | JavaScript       | Minecraft variant built on tetrahedral geometry — voxels replaced by Quadray-coordinate tetrahedra, exploring Buckminster Fuller's synergetics in an interactive 3D world                                                                                                                                                                                                                             |
| <b>docxology/MetaInferAnt</b><br><a href="https://github.com/docxology/MetaInferAnt">https://github.com/docxology/MetaInferAnt</a>                                                          | Python           | Meta-framework integrating computational entomology, Active Inference, and information theory for modeling ant colony cognition and beyond<br><br>paper: papers/2021_ActiveInferants/                                                                                                                                                                                                                 |
| <b>docxology/codomymrnx</b><br><a href="https://github.com/docxology/codomymrnx">https://github.com/docxology/codomymrnx</a>                                                                | Python           | AI-native modular coding workspace — 100+ modules, 500+ MCP tools, 21K+ zero-mock tests, multi-agent orchestration (Claude/Gemini/GPT), PAI integration                                                                                                                                                                                                                                               |
| <b>docxology/cognitive-engine</b><br><a href="https://github.com/docxology/cognitive-engine">https://github.com/docxology/cognitive-engine</a>                                              | Python           | Hybrid neuro-symbolic AI combining fractal symbolic representations with probabilistic neural models for enhanced reasoning; includes Unipixel fractal building blocks and PEFf ethical reasoning                                                                                                                                                                                                     |
| <b>docxology/opentir</b><br><a href="https://github.com/docxology/opentir">https://github.com/docxology/opentir</a>                                                                         | Python           | Comprehensive toolkit analyzing 250+ Palantir open-source repositories (6.2M+ lines of code) — automated discovery, multi-language analysis, dependency mapping, statistical profiling, and professional visualizations                                                                                                                                                                               |
| <b>docxology/timeline_generator</b><br><a href="https://github.com/docxology/timeline_generator">https://github.com/docxology/timeline_generator</a>                                        | TypeScript       | Networked Life Encoding — interactive D3 knowledge graph of human intellectual history; 24 relationship types, Perplexity AI enrichment; seeded around R. Buckminster Fuller's network                                                                                                                                                                                                                |
| <b>docxology/FORMINDEX</b><br><a href="https://github.com/docxology/FORMINDEX">https://github.com/docxology/FORMINDEX</a>                                                                   | Rich Text Format | Analysis and tooling around FORMIS, the world's largest ant-literature database; bibliometric mining and structured index generation                                                                                                                                                                                                                                                                  |
| <b>docxology/ActiveInferAnts</b><br><a href="https://github.com/docxology/ActiveInferAnts">https://github.com/docxology/ActiveInferAnts</a>                                                 | Python           | Active Inference for ants — multi-language implementations (Python) for modeling ant-colony cognition as Bayesian agents minimizing free energy<br><br>paper: papers/2021_ActiveInferants/                                                                                                                                                                                                            |
| <b>docxology/crescent-city</b><br><a href="https://github.com/docxology/crescent-city">https://github.com/docxology/crescent-city</a>                                                       | TypeScript       | End-to-end pipeline for ingesting, parsing, and querying the Crescent City, CA municipal code; TypeScript-based legal-text processing                                                                                                                                                                                                                                                                 |
| <b>docxology/ivm-xyz</b><br><a href="https://github.com/docxology/ivm-xyz">https://github.com/docxology/ivm-xyz</a>                                                                         | Python           | Python library for 3D and 4D Quadray/IVM (Isotropic Vector Matrix) geometry — coordinate conversion, volume calculations, and polyhedral operations                                                                                                                                                                                                                                                   |
| <b>docxology/Markov_Blanket_Detection</b><br><a href="https://github.com/docxology/Markov_Blanket_Detection">https://github.com/docxology/Markov_Blanket_Detection</a>                      | Python           | Dynamic Markov Blanket Detection (DMBD) — PyTorch-accelerated framework for identifying causal relationships in complex systems; static and temporal modes with cognitive-structure labeling                                                                                                                                                                                                          |
| <b>docxology/QuadMath</b><br><a href="https://github.com/docxology/QuadMath">https://github.com/docxology/QuadMath</a>                                                                      | TeX              | Mathematical exposition of Quadray coordinate systems (4-vector tetrahedral coordinates); LaTeX documentation and computation library                                                                                                                                                                                                                                                                 |
| <b>docxology/active_inference</b><br><a href="https://github.com/docxology/active_inference">https://github.com/docxology/active_inference</a>                                              | Python           | Active Inference for, with, and by Generative AI — Python implementations of free-energy-minimizing agents integrated with modern AI tooling<br><br>paper: papers/2020_BehaviorEngineering/                                                                                                                                                                                                           |
| <b>docxology/AgenticMesh</b><br><a href="https://github.com/docxology/AgenticMesh">https://github.com/docxology/AgenticMesh</a>                                                             | Python           | Active Inference & Agentic Mesh — modular Python infrastructure for building agentic systems using free-energy-minimizing Active Inference principles                                                                                                                                                                                                                                                 |
| <b>docxology/curriculum</b><br><a href="https://github.com/docxology/curriculum">https://github.com/docxology/curriculum</a>                                                                | HTML             | Curriculum development framework for interdisciplinary education in Active Inference, biology, and computational science                                                                                                                                                                                                                                                                              |
| <b>docxology/lean_niche</b><br><a href="https://github.com/docxology/lean_niche">https://github.com/docxology/lean_niche</a>                                                                | Python           | Lean/minimal niche-construction modeling — formal computational representations of ecological niche dynamics                                                                                                                                                                                                                                                                                          |
| <b>docxology/literature</b><br><a href="https://github.com/docxology/literature">https://github.com/docxology/literature</a>                                                                | Python           | Literature collection and analysis tools for structured ingestion, annotation, and exploration of scientific publications                                                                                                                                                                                                                                                                             |
| <b>docxology/obsidian-construction-from-text</b><br><a href="https://github.com/docxology/obsidian-construction-from-text">https://github.com/docxology/obsidian-construction-from-text</a> | Python           | Automated Obsidian vault construction from unstructured text — NLP pipelines to generate bidirectionally-linked knowledge graphs                                                                                                                                                                                                                                                                      |
| <b>docxology/qv_live_protocol</b><br><a href="https://github.com/docxology/qv_live_protocol">https://github.com/docxology/qv_live_protocol</a>                                              | Python           | QR code-based live protocol system for real-time data exchange and interactive session management                                                                                                                                                                                                                                                                                                     |



| Repository                                                                                                                                                                                                                                                                      | Stack  | Description and Source Links                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>docxology/AGEINT</b><br><a href="https://github.com/docxology/AGEINT">https://github.com/docxology/AGEINT</a>                                                                                                                                                                | HTML   | Agentic Intelligence curriculum — modular educational scaffold for teaching bounded AI-agent support and synthetic analytic tradecraft · · Zenodo<br>paper: papers/2026_AGEINT/   zenodo: <a href="https://doi.org/10.5281/zenodo.20732274">https://doi.org/10.5281/zenodo.20732274</a>                      |
| <b>docxology/template_autoscientists</b><br><a href="https://github.com/docxology/template_autoscientists">https://github.com/docxology/template_autoscientists</a>                                                                                                             | Python | AutoScientists deterministic multi-agent scientific-discovery coordination harness · · Zenodo<br>paper: papers/2026_DeterministicTestbedSelf/   zenodo: <a href="https://doi.org/10.5281/zenodo.20533669">https://doi.org/10.5281/zenodo.20533669</a>                                                        |
| <b>docxology/template_newspaper</b><br><a href="https://github.com/docxology/template_newspaper">https://github.com/docxology/template_newspaper</a>                                                                                                                            | Python | The Triplicate data-driven large-format newspaper layout engine · · Zenodo<br>paper: papers/2026_Triplicate/   zenodo: <a href="https://doi.org/10.5281/zenodo.20533675">https://doi.org/10.5281/zenodo.20533675</a>                                                                                         |
| <b>docxology/ntqr_llm</b><br><a href="https://github.com/docxology/ntqr_llm">https://github.com/docxology/ntqr_llm</a>                                                                                                                                                          | Python | Algebraic NTQR evaluation study recovering LLM-persona accuracies from unlabeled votes · · Zenodo<br>paper: papers/2026_RecoveringLLMPersona/   zenodo: <a href="https://doi.org/10.5281/zenodo.20498699">https://doi.org/10.5281/zenodo.20498699</a>                                                        |
| <b>docxology/itrace</b><br><a href="https://github.com/docxology/itrace">https://github.com/docxology/itrace</a>                                                                                                                                                                | Python | Verification-first webcam eye-movement analysis toolkit for gaze, saccades, pupillometry, and diagnostic-quality exports · · Zenodo<br>paper: papers/2026_ITrace/   zenodo: <a href="https://doi.org/10.5281/zenodo.20614908">https://doi.org/10.5281/zenodo.20614908</a>                                    |
| <b>docxology/coasys</b><br><a href="https://github.com/docxology/coasys">https://github.com/docxology/coasys</a>                                                                                                                                                                | Python | Integration workspace aggregating and exercising the coasys (AD4M) agent-centric social-layer repositories                                                                                                                                                                                                   |
| <b>docxology/CogSecSkills</b><br><a href="https://github.com/docxology/CogSecSkills">https://github.com/docxology/CogSecSkills</a>                                                                                                                                              | Python | Multiharness cognitive-security and analytic-tradecraft skill library — 100 defensive SAT/cogsec skills runnable under Claude Code, Codex, and Hermes · · Zenodo<br>paper: papers/2026_CogSecSkills/   zenodo: <a href="https://doi.org/10.5281/zenodo.20804585">https://doi.org/10.5281/zenodo.20804585</a> |
| <b>docxology/template_madlib</b><br><a href="https://github.com/docxology/template_madlib">https://github.com/docxology/template_madlib</a>                                                                                                                                     | Python | Deterministic token-injection exemplar for conditional IMRAD manuscripts — reproducible madlib-style variable substitution for scientific writing · · Zenodo<br>paper: papers/2026_TemplateMadlib/   zenodo: <a href="https://doi.org/10.5281/zenodo.20786638">https://doi.org/10.5281/zenodo.20786638</a>   |
| <b>docxology/ultralink-docx</b><br><a href="https://github.com/docxology/ultralink-docx">https://github.com/docxology/ultralink-docx</a>                                                                                                                                        | HTML   | UltraLink document format tooling — linking, rendering, and transformation utilities for rich hyperlinked documents                                                                                                                                                                                          |
| <b>docxology/cognitive</b><br><a href="https://github.com/docxology/cognitive">https://github.com/docxology/cognitive</a>                                                                                                                                                       | Python | Cognitive Ecosystem Modeling Framework — Active Inference agents with Obsidian-compatible knowledge management, bidirectional graph validation, belief updating, and network visualization                                                                                                                   |
| <b>docxology/active_torchference</b><br><a href="https://github.com/docxology/active_torchference">https://github.com/docxology/active_torchference</a>                                                                                                                         | Python | PyTorch-based Active Inference implementations — GPU-accelerated variational inference and free energy minimization                                                                                                                                                                                          |
| <b>docxology/ant-pheromone</b><br><a href="https://github.com/docxology/ant-pheromone">https://github.com/docxology/ant-pheromone</a>                                                                                                                                           | Python | Computational simulation of ant pheromone trail dynamics and stigmergic communication                                                                                                                                                                                                                        |
| <b>docxology/ant_stack</b><br><a href="https://github.com/docxology/ant_stack">https://github.com/docxology/ant_stack</a>                                                                                                                                                       | Python | The Ant Stack — layered computational model of collective ant-colony intelligence<br>paper: papers/2025_AntStack/                                                                                                                                                                                            |
| <b>docxology/cohereants</b><br><a href="https://github.com/docxology/cohereants">https://github.com/docxology/cohereants</a>                                                                                                                                                    | Python | Computational research on insect infrared sensing and olfaction modeling · · Zenodo · Zenodo<br>paper: papers/2026_WhenDoBugs/   zenodo: <a href="https://doi.org/10.5281/zenodo.20450970">https://doi.org/10.5281/zenodo.20450970</a>                                                                       |
| <b>docxology/ento_linguistics</b><br><a href="https://github.com/docxology/ento_linguistics">https://github.com/docxology/ento_linguistics</a>                                                                                                                                  | Python | Ento-Linguistics corpus pipeline — term extraction, terminology networks, semantic entropy, CACE scoring<br>paper: papers/2026_EntoLinguistics/                                                                                                                                                              |
| <b>docxology/cognitive_case_diagrams</b><br><a href="https://github.com/docxology/cognitive_case_diagrams">https://github.com/docxology/cognitive_case_diagrams</a>                                                                                                             | TeX    | Compositional linguistic case for cognitive modeling — categorical case systems, DisCoCat/DisCoCirc diagrams, tests and figures for the Active Inference Journal manuscript<br>paper: papers/2026_CognitiveCaseDiagrams/                                                                                     |
| <b>docxology/Autonomous-Drone-Navigation-in-AirSim-With-Active-Inference</b><br><a href="https://github.com/docxology/Autonomous-Drone-Navigation-in-AirSim-With-Active-Inference">https://github.com/docxology/Autonomous-Drone-Navigation-in-AirSim-With-Active-Inference</a> | Julia  | Active Inference-driven autonomous drone navigation in Microsoft AirSim simulation environment                                                                                                                                                                                                               |
| <b>docxology/biol-1</b><br><a href="https://github.com/docxology/biol-1">https://github.com/docxology/biol-1</a>                                                                                                                                                                | HTML   | BIOL-1: General Biology course materials — College of the Redwoods; lectures, labs, and interactive HTML content                                                                                                                                                                                             |
| <b>docxology/biol-8</b><br><a href="https://github.com/docxology/biol-8">https://github.com/docxology/biol-8</a>                                                                                                                                                                | HTML   | BIOL-8: Human Biology course materials — College of the Redwoods; lectures, labs, and interactive HTML content                                                                                                                                                                                               |





| Repository                                                                                                                                                                      | Stack       | Description and Source Links                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>docxology/biology_textbook</b><br><a href="https://github.com/docxology/biology_textbook">https://github.com/docxology/biology_textbook</a>                                  | Python      | Open generative biology textbook — Markdown source, tested Python modules, programmatic figures; archived at Zenodo<br>paper: papers/2026_BiologyTextbook/   zenodo: <a href="https://doi.org/10.5281/zenodo.20286478">https://doi.org/10.5281/zenodo.20286478</a> |
| <b>docxology/cascadia</b><br><a href="https://github.com/docxology/cascadia">https://github.com/docxology/cascadia</a>                                                          | Python      | Cascadia bioregion modeling and data analysis tools                                                                                                                                                                                                                |
| <b>docxology/course</b><br><a href="https://github.com/docxology/course">https://github.com/docxology/course</a>                                                                | Python      | Course materials framework for structured interdisciplinary learning modules                                                                                                                                                                                       |
| <b>docxology/enactive_inference_model</b><br><a href="https://github.com/docxology/enactive_inference_model">https://github.com/docxology/enactive_inference_model</a>          | Python      | Three-level hierarchical enactive inference model of mental action — focused-attention meditation with expert/novice profiles; reproducible simulations and figures<br>paper: papers/2025_Thoughtseeds/                                                            |
| <b>docxology/EvoJump</b><br><a href="https://github.com/docxology/EvoJump">https://github.com/docxology/EvoJump</a>                                                             | Python      | Comprehensive Framework for Evolutionary Ontogenetic Analysis — modeling developmental trajectories and evolutionary dynamics<br>paper: papers/2025_EvoJump/                                                                                                       |
| <b>docxology/flick</b><br><a href="https://github.com/docxology/flick">https://github.com/docxology/flick</a>                                                                   | Python      | Lightweight Python tooling for rapid data flicking and processing pipelines                                                                                                                                                                                        |
| <b>docxology/fuller-obsidian</b><br><a href="https://github.com/docxology/fuller-obsidian">https://github.com/docxology/fuller-obsidian</a>                                     | unspecified | Obsidian vault dedicated to R. Buckminster Fuller's ideas — synergetics, geodesics, and design science                                                                                                                                                             |
| <b>docxology/godel_ivm</b><br><a href="https://github.com/docxology/godel_ivm">https://github.com/docxology/godel_ivm</a>                                                       | Python      | Gödel numbering meets the IVM (Isotropic Vector Matrix) — formal encoding of geometric objects as prime products                                                                                                                                                   |
| <b>docxology/goference</b><br><a href="https://github.com/docxology/goference">https://github.com/docxology/goference</a>                                                       | Go          | Active Goference — production-ready Go framework for Active Inference autonomous agents; genuine variational inference, free energy minimization, POMDP policy planning via gonum                                                                                  |
| <b>docxology/hhs-opendata</b><br><a href="https://github.com/docxology/hhs-opendata">https://github.com/docxology/hhs-opendata</a>                                              | Python      | Analysis of HHS (Department of Health and Human Services) open data — policy analytics and public-health data pipelines                                                                                                                                            |
| <b>docxology/grateful_data</b><br><a href="https://github.com/docxology/grateful_data">https://github.com/docxology/grateful_data</a>                                           | Python      | The Music Never Stopped citation-bound Grateful Dead data compendium · · Zenodo ·<br>paper: papers/2026_MusicNeverStopped/   zenodo: <a href="https://doi.org/10.5281/zenodo.20482026">https://doi.org/10.5281/zenodo.20482026</a>                                 |
| <b>docxology/infra-calc</b><br><a href="https://github.com/docxology/infra-calc">https://github.com/docxology/infra-calc</a>                                                    | JavaScript  | Infrastructure calculator for estimating resource requirements and costs in multi-agent and AI systems                                                                                                                                                             |
| <b>docxology/markdown_decision_processes</b><br><a href="https://github.com/docxology/markdown_decision_processes">https://github.com/docxology/markdown_decision_processes</a> | Python      | Framework for Probabilistic Document Analysis — Markov Decision Process over Markdown artifacts for structured document reasoning<br>paper: papers/2025_MarkdownDecisionProcess/                                                                                   |
| <b>docxology/mdkv</b><br><a href="https://github.com/docxology/mdkv">https://github.com/docxology/mdkv</a>                                                                      | Python      | MKV-like key-value format and methods for Markdown — structured metadata embedding in plain-text documents<br>paper: papers/2025_MDKV/                                                                                                                             |
| <b>docxology/p3if</b><br><a href="https://github.com/docxology/p3if">https://github.com/docxology/p3if</a>                                                                      | Python      | Properties, Processes, and Perspectives Inter-Framework (P3IF) — multiplexing interdisciplinary requirements frameworks to manage information risk and foster cognitive security<br>paper: papers/2023_P3IF/                                                       |
| <b>docxology/service</b><br><a href="https://github.com/docxology/service">https://github.com/docxology/service</a>                                                             | Python      | Service-layer infrastructure and API tooling for Python-based research applications                                                                                                                                                                                |
| <b>docxology/snake</b><br><a href="https://github.com/docxology/snake">https://github.com/docxology/snake</a>                                                                   | Python      | Classic snake game implemented in Python — used for testing multi-agent and reinforcement learning algorithms                                                                                                                                                      |
| <b>docxology/steganographer</b><br><a href="https://github.com/docxology/steganographer">https://github.com/docxology/steganographer</a>                                        | Rust        | High-performance Rust tool for embedding BLAKE3+Ed25519 cryptographic signatures and visible watermarks into live video/audio via LSB steganography; four Rust crates, 132 tests, GStreamer integration                                                            |
| <b>docxology/template_textbook</b><br><a href="https://github.com/docxology/template_textbook">https://github.com/docxology/template_textbook</a>                               | HTML        | Modular fillable scaffold for book-length technical works · · Zenodo<br>paper: papers/2026_TemplateTextbook/   zenodo: <a href="https://doi.org/10.5281/zenodo.20533125">https://doi.org/10.5281/zenodo.20533125</a>                                               |
| <b>docxology/blake_jiang</b><br><a href="https://github.com/docxology/blake_jiang">https://github.com/docxology/blake_jiang</a>                                                 | Python      | William Blake, Professor Jiang, and the Architecture of Intelligence — AI-augmented scholarly response to Game Theory #24 with Blake, Jiang, and Active Inference framing<br>paper: papers/2026_BlakeJiang/                                                        |



| Repository                                                                                                                                                                                                                         | Stack       | Description and Source Links                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>docxology/symergetics</b><br><a href="https://github.com/docxology/symergetics">https://github.com/docxology/symergetics</a>                                                                                                    | Python      | Symbolic Synergetics — computer algebra and formal symbolic systems for Buckminster Fuller's synergetics mathematics                                                                                                                                                                                                    |
| <b>docxology/transformer</b><br><a href="https://github.com/docxology/transformer">https://github.com/docxology/transformer</a>                                                                                                    | Python      | Transformer architecture explorations — attention mechanisms and architectural variants for Active Inference research                                                                                                                                                                                                   |
| <b>ActiveInferenceInstitute/ActiveInferenceJournal</b><br><a href="https://github.com/ActiveInferenceInstitute/ActiveInferenceJournal">https://github.com/ActiveInferenceInstitute/ActiveInferenceJournal</a>                      | HTML        | Content hub for the Active Inference Journal — 500+ video transcripts, stream archives, and multimedia publishing infrastructure for the Institute's primary educational output                                                                                                                                         |
| <b>ActiveInferenceInstitute/ActiveBlockference</b><br><a href="https://github.com/ActiveInferenceInstitute/ActiveBlockference">https://github.com/ActiveInferenceInstitute/ActiveBlockference</a>                                  | Jupyter     | Active Inference agents in cadCAD block-based simulation — formal agent-based modeling using complex-systems simulation framework                                                                                                                                                                                       |
| <b>ActiveInferenceInstitute/ActiveInferAnts</b><br><a href="https://github.com/ActiveInferenceInstitute/ActiveInferAnts">https://github.com/ActiveInferenceInstitute/ActiveInferAnts</a>                                           | Python      | Active Inference models for/of ants across 32+ programming languages — multi-language reference implementation of stigmergic Active Inference                                                                                                                                                                           |
| <b>ActiveInferenceInstitute/GeneralizedNotationNotation</b><br><a href="https://github.com/ActiveInferenceInstitute/GeneralizedNotationNotation">https://github.com/ActiveInferenceInstitute/GeneralizedNotationNotation</a>       | Python      | GNN — formal notation standard for specifying Active Inference generative models; JSON/YAML schema, validators, and transpilers to Python, Julia, and MATLAB · Zenodo<br>zenodo: <a href="https://doi.org/10.5281/zenodo.19600217">https://doi.org/10.5281/zenodo.19600217</a>                                          |
| <b>ActiveInferenceInstitute/fep_lean</b><br><a href="https://github.com/ActiveInferenceInstitute/fep_lean">https://github.com/ActiveInferenceInstitute/fep_lean</a>                                                                | Python      | Lean 4 / Mathlib4 formalization of FEP-related mathematics — 50-topic sorry-free catalog, Hermes/OpenGauss LLM drafting, zero-mock verification, manuscript pipeline · · Zenodo<br>paper: papers/2026_FEPLearn/   zenodo: <a href="https://doi.org/10.5281/zenodo.19699234">https://doi.org/10.5281/zenodo.19699234</a> |
| <b>ActiveInferenceInstitute/cognitive</b><br><a href="https://github.com/ActiveInferenceInstitute/cognitive">https://github.com/ActiveInferenceInstitute/cognitive</a>                                                             | Python      | Cognitive modeling and simulation tools — multi-agent Active Inference environments, benchmarks, and visualization                                                                                                                                                                                                      |
| <b>ActiveInferenceInstitute/Active_Inference_Ontology</b><br><a href="https://github.com/ActiveInferenceInstitute/Active_Inference_Ontology">https://github.com/ActiveInferenceInstitute/Active_Inference_Ontology</a>             | unspecified | Formal OWL/RDF ontology for Active Inference — defines classes, properties, and axioms for free energy principle concepts                                                                                                                                                                                               |
| <b>ActiveInferenceInstitute/Journal-Utilities</b><br><a href="https://github.com/ActiveInferenceInstitute/Journal-Utilities">https://github.com/ActiveInferenceInstitute/Journal-Utilities</a>                                     | HTML        | Transcription, processing, and publishing pipelines for the Active Inference Journal — Whisper-based download, transcription, RAG/knowledge-graph, and export utilities · Zenodo release<br>zenodo: <a href="https://doi.org/10.5281/zenodo.18686966">https://doi.org/10.5281/zenodo.18686966</a>                       |
| <b>ActiveInferenceInstitute/Symposium</b><br><a href="https://github.com/ActiveInferenceInstitute/Symposium">https://github.com/ActiveInferenceInstitute/Symposium</a>                                                             | Python      | Applied Active Inference Symposium materials — talks, workshops, proceedings, and collaborative session artifacts from annual Applied AIF symposia                                                                                                                                                                      |
| <b>ActiveInferenceInstitute/ActiveInferenceCategoryTheory</b><br><a href="https://github.com/ActiveInferenceInstitute/ActiveInferenceCategoryTheory">https://github.com/ActiveInferenceInstitute/ActiveInferenceCategoryTheory</a> | unspecified | Formal categorical foundations for Active Inference — string diagrams, functorial semantics, and compositional models using category theory                                                                                                                                                                             |
| <b>ActiveInferenceInstitute/CEREBRUM</b><br><a href="https://github.com/ActiveInferenceInstitute/CEREBRUM">https://github.com/ActiveInferenceInstitute/CEREBRUM</a>                                                                | Python      | Case-Enabled Reasoning Engine with Bayesian Representations for Unified Modeling — grammatical case framework for compositional cognitive models · Zenodo<br>paper: papers/2025_CEREBRUM/   zenodo: <a href="https://doi.org/10.5281/zenodo.15170907">https://doi.org/10.5281/zenodo.15170907</a>                       |
| <b>ActiveInferenceInstitute/GEO-INFER</b><br><a href="https://github.com/ActiveInferenceInstitute/GEO-INFER">https://github.com/ActiveInferenceInstitute/GEO-INFER</a>                                                             | HTML        | Geospatial Active Inference — applying free energy minimization to geographic systems, spatial planning, and bioregional intelligence                                                                                                                                                                                   |
| <b>ActiveInferenceInstitute/AEOS</b><br><a href="https://github.com/ActiveInferenceInstitute/AEOS">https://github.com/ActiveInferenceInstitute/AEOS</a>                                                                            | unspecified | Active Entity Ontology for Science — OWL ontology for representing active entities, their actions, and causal relationships in scientific models                                                                                                                                                                        |
| <b>ActiveInferenceInstitute/courses</b><br><a href="https://github.com/ActiveInferenceInstitute/courses">https://github.com/ActiveInferenceInstitute/courses</a>                                                                   | HTML        | Structured Active Inference learning courses — cohort-based curriculum materials, exercises, and collaborative learning artifacts                                                                                                                                                                                       |
| <b>ActiveInferenceInstitute/Research-Discovery-Engine</b><br><a href="https://github.com/ActiveInferenceInstitute/Research-Discovery-Engine">https://github.com/ActiveInferenceInstitute/Research-Discovery-Engine</a>             | HTML        | AI-driven research synthesis and navigation — semantic search, citation graph analysis, and cross-domain connection discovery across the Active Inference literature                                                                                                                                                    |





| Repository                                                                                                                                                                                                                               | Stack       | Description and Source Links                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ActiveInferenceInstitute/Parr_et_al_2022_ActInf_Textbook</b><br><a href="https://github.com/ActiveInferenceInstitute/Parr_et_al_2022_ActInf_Textbook">https://github.com/ActiveInferenceInstitute/Parr_et_al_2022_ActInf_Textbook</a> | Julia       | Active Inference Textbook Study Group — code, exercises, and reading notes accompanying Parr, Pezzulo & Friston (2022)                                                                                                                                                                                                                                            |
| <b>ActiveInferenceInstitute/Biofirm</b><br><a href="https://github.com/ActiveInferenceInstitute/Biofirm">https://github.com/ActiveInferenceInstitute/Biofirm</a>                                                                         | Python      | Bioregional Active Inference for organizational stewardship — applying free energy minimization to ecological and community governance systems                                                                                                                                                                                                                    |
| <b>ActiveInferenceInstitute/Start</b><br><a href="https://github.com/ActiveInferenceInstitute/Start">https://github.com/ActiveInferenceInstitute/Start</a>                                                                               | Python      | Institute onboarding hub — getting-started guides, contributor pathways, project index, and orientation materials for new participants                                                                                                                                                                                                                            |
| <b>ActiveInferenceInstitute/Knowledge-Engineering</b><br><a href="https://github.com/ActiveInferenceInstitute/Knowledge-Engineering">https://github.com/ActiveInferenceInstitute/Knowledge-Engineering</a>                               | Jupyter     | Knowledge engineering tools and workflows — ontology alignment, concept mapping, and structured knowledge-base construction for Active Inference                                                                                                                                                                                                                  |
| <b>ActiveInferenceInstitute/ActInf_RxInfer</b><br><a href="https://github.com/ActiveInferenceInstitute/ActInf_RxInfer">https://github.com/ActiveInferenceInstitute/ActInf_RxInfer</a>                                                    | unspecified | Active Inference with RxInfer.jl — Julia-based reactive message passing for efficient Bayesian inference in Active Inference models                                                                                                                                                                                                                               |
| <b>ActiveInferenceInstitute/pymcp</b><br><a href="https://github.com/ActiveInferenceInstitute/pymcp">https://github.com/ActiveInferenceInstitute/pymcp</a>                                                                               | Python      | PyMCP — Model Context Protocol bridge for PyMDP, enabling Active Inference agents to interoperate with MCP-compatible AI tooling                                                                                                                                                                                                                                  |
| <b>ActiveInferenceInstitute/GEN24</b><br><a href="https://github.com/ActiveInferenceInstitute/GEN24">https://github.com/ActiveInferenceInstitute/GEN24</a>                                                                               | Jupyter     | GEN-24 working project — collaborative generative systems research from the 2024 active inference cohort                                                                                                                                                                                                                                                          |
| <b>ActiveInferenceInstitute/NoOrg</b><br><a href="https://github.com/ActiveInferenceInstitute/NoOrg">https://github.com/ActiveInferenceInstitute/NoOrg</a>                                                                               | HTML        | NoOrg is a Real Org — experimental organizational infrastructure and governance frameworks for decentralized scientific institutions                                                                                                                                                                                                                              |
| <b>ActiveInferenceInstitute/ultralink</b><br><a href="https://github.com/ActiveInferenceInstitute/ultralink">https://github.com/ActiveInferenceInstitute/ultralink</a>                                                                   | unspecified | UltraLink — framework for linking, rendering, and transforming richly cross-referenced documents across multiple output formats                                                                                                                                                                                                                                   |
| <b>ActiveInferenceInstitute/ActiveInferenceImplementations</b><br><a href="https://github.com/ActiveInferenceInstitute/ActiveInferenceImplementations">https://github.com/ActiveInferenceInstitute/ActiveInferenceImplementations</a>    | Jupyter     | Learning and exploring probabilistic inference — beginner-accessible Jupyter notebooks for Active Inference from first principles                                                                                                                                                                                                                                 |
| <b>ActiveInferenceInstitute/ATLAS</b><br><a href="https://github.com/ActiveInferenceInstitute/ATLAS">https://github.com/ActiveInferenceInstitute/ATLAS</a>                                                                               | Python      | Question-Oriented Pattern Languages for Knowledge Management — structured templates for navigating complex scientific knowledge spaces                                                                                                                                                                                                                            |
| <b>ActiveInferenceInstitute/COGANT</b><br><a href="https://github.com/ActiveInferenceInstitute/COGANT">https://github.com/ActiveInferenceInstitute/COGANT</a>                                                                            | Python      | Deterministic Codebase-to-GNN Translation — evidence compiler converting repositories into Generalized Notation Notation Active Inference artifacts via a 22-rule fixpoint engine; dual Python/Rust · · Zenodo · Zenodo<br><br>paper: papers/2026_COGANT/   zenodo: <a href="https://doi.org/10.5281/zenodo.20705351">https://doi.org/10.5281/zenodo.20705351</a> |
| <b>ActiveInferenceInstitute/on_policy_distillation</b><br><a href="https://github.com/ActiveInferenceInstitute/on_policy_distillation">https://github.com/ActiveInferenceInstitute/on_policy_distillation</a>                            | Python      | On-Policy Distillation and Active Inference — finite variational-model correspondence with deterministic witnesses, reproducibility artifacts, and paired publication release · · Zenodo · Zenodo<br><br>paper: papers/2026_PolicyDistillationAs/   zenodo: <a href="https://doi.org/10.5281/zenodo.20749817">https://doi.org/10.5281/zenodo.20749817</a>         |
| <b>ActiveInferenceInstitute/Bots</b><br><a href="https://github.com/ActiveInferenceInstitute/Bots">https://github.com/ActiveInferenceInstitute/Bots</a>                                                                                  | unspecified | Institute bot infrastructure — automation, notification, and community management bots for the Active Inference Institute's platforms                                                                                                                                                                                                                             |
| <b>ActiveInferenceInstitute/Garden-of-Iris</b><br><a href="https://github.com/ActiveInferenceInstitute/Garden-of-Iris">https://github.com/ActiveInferenceInstitute/Garden-of-Iris</a>                                                    | Python      | Interactive visualization garden — generative art and data-visualization experiments exploring Active Inference dynamics                                                                                                                                                                                                                                          |
| <b>ActiveInferenceInstitute/gen25</b><br><a href="https://github.com/ActiveInferenceInstitute/gen25">https://github.com/ActiveInferenceInstitute/gen25</a>                                                                               | unspecified | GEN-25 working project — generative systems and collaborative research from the 2025 active inference cohort                                                                                                                                                                                                                                                      |
| <b>ActiveInferenceInstitute/gnn</b><br><a href="https://github.com/ActiveInferenceInstitute/gnn">https://github.com/ActiveInferenceInstitute/gnn</a>                                                                                     | unspecified | Generalized Notation Notation v2 — next-generation GNN specification with extended model classes and improved toolchain                                                                                                                                                                                                                                           |
| <b>ActiveInferenceInstitute/ISSS</b><br><a href="https://github.com/ActiveInferenceInstitute/ISSS">https://github.com/ActiveInferenceInstitute/ISSS</a>                                                                                  | Python      | International Society for the Systems Sciences collaboration — joint research and educational materials bridging Active Inference and systems science                                                                                                                                                                                                             |
| <b>ActiveInferenceInstitute/topp</b><br><a href="https://github.com/ActiveInferenceInstitute/topp">https://github.com/ActiveInferenceInstitute/topp</a>                                                                                  | TeX         | The Open Problems Project — curated registry of open scientific problems in Active Inference and the free energy principle                                                                                                                                                                                                                                        |



## Conferences, Posters, Seminars, Workshops (17)

### 2025 Inference about Foraging and Foraging as Inference | Queens College, CUNY, Biology

- Talk

Source: Queens College Biology <https://www.qc.cuny.edu/academics/bio/>

### 2025 Systems Processes, Active Inference, and Beyond | Enduring Patterns, Emerging Futures: Celebrating Dr. Len Troncale

- Talk 2025-09-17

Source: Zenodo presentation <https://doi.org/10.5281/zenodo.17138223>

### 2024 ...And Eternity in an hour of Systems Science Education and Knowledge Engineering | ISSS 2024

- Workshop

Source: ISSS <https://isss.org/world/>

### 2024 Funding AI public goods and keeping them public | Funding the Commons

- Panel

Source: Funding the Commons <https://fundingthecommons.io/>

### 2024 Intelligence in the Age of LLMs | European Identity and Cloud Conference 2024

- Panel

Source: KuppingerCole EIC speaker <https://www.kuppingercole.com/events/eic2024/speakers/3516>

### 2022 Tissue-specific gene expression meta-analysis in honey bees (*Apis mellifera*) | Entomology Society of America, Pacific Branch

- Talk 2022-04-10

Source: Entomological Society of America <https://entsoc.org/events>

### 2021 Thinking like a State: Active inference and the deep roots of complex societies | Cognition 2021

- Talk Avel Guenin-Carlut, Daniel Friedman, and Virginia Bleu Knight.

Source: Cognition <https://cognition.uqam.ca/>

### 2021 Introducing Active Inference as An Integrative Principle in The Study of Collective Intelligence | ACM-CI 2021

- Talk Avel Guenin-Carlut and Daniel Friedman.

Source: ACM Collective Intelligence <https://ci.acm.org/>

### 2020 Active Inference for Communicating Systems | IIBM, Chile

- Talk Invited online seminar. 2020-11-03

Source: IIBM <https://www.iibm.cl/>

### 2019 Invited seminar, Department of Animal Behavior, University of California, Davis | Essig Brunch Seminar Series

- Talk 2019-04-05

Source: UC Davis Entomology <https://entomology.ucdavis.edu/>

### 2018 Invited seminar, SUMS-RAS | Stanford University Mass Spectrometry Research Symposium

- Talk 2018-10-04

Source: Stanford Mass Spectrometry <https://mass-spec.stanford.edu/>

### 2018 Invited seminar, University of California, Berkeley | Essig Entomology Brunch

- Talk 2018-02-06

Source: Essig Museum <https://essig.berkeley.edu/>

### 2017 The role of dopamine in the regulation of foraging in the red harvester ant | Entomology 2017

- Poster

Source: Entomological Society of America <https://entsoc.org/events/annual-meeting>

### 2017 Quantitative LC-MS/MS analysis of dopamine in single ant brain samples | ASMS 2017

- Poster KM Krasinska, DA Friedman, DM Gordon, AS Chien.

Source: ASMS Annual Conference <https://www.asms.org/conferences/annual-conference>



- 2015 Transcriptomic analysis of the regulation of foraging in harvester ants | Biology and Genetics of Social Insects, CSHL 2015**  
 - Poster DA Friedman, A Pilko, DM Gordon.  
 Source: Cold Spring Harbor Meetings <https://meetings.cshl.edu/>
- 2014 Evolution of sex comb enhancers in the HOX gene Scr | Drosophila Research Conference 2014**  
 - Poster DA Friedman, T Orenic, C McCullough, E Eksi, OY Barmina, A Kopp.  
 Source: GSA Drosophila Conference <https://genetics-gsa.org/drosophila/>
- 2012 Visualizing the Integration of Sensory Neurons into a Sex-Specific Central Nervous System Circuit in 3-D | UC Davis Undergraduate Research Conference**  
 - Talk DA Friedman.  
 Source: UC Davis Undergraduate Research Center <https://urc.ucdavis.edu/>

## Science Engagement, Outreach, Quotes, Articles (29)

- 2026 S3E10 The FEP: A Tool for Understanding and Modelling Complex Systems**  
 - Video conversation on another channel  
 Source: BathmikeC podcast <https://bathmikec.podbean.com/e/s3e10-the-fep-a-tool-for-understanding-and-modelling-complex-systems/>
- 2025 (DIA 1/SESSAO 1) Lancamento do Livro: As Colonias de Formigas Sao Conscientes?**  
 - Video conversation on another channel  
 Source: YouTube <https://www.youtube.com/watch?v=BJ-PAT3cgf4>
- 2024 From Ants to Active Inference with Daniel Ari Friedman**  
 - Video conversation on another channel | Love and Philosophy, Andrea Hiott  
 Source: YouTube [https://www.youtube.com/watch?v=x\\_VPw1K55BM](https://www.youtube.com/watch?v=x_VPw1K55BM)
- 2024 Building Smarter AI: Active Inference and Nested Models**  
 - Video conversation on another channel | Bagaev, de Vries, Friedman, Pashea.  
 Source: YouTube <https://youtu.be/u3w24FtAsKw>
- 2024 Interaction. In Search of Daniel Friedman's Deepest Value**  
 - Video conversation on another channel | Math4Wisdom  
 Source: YouTube <https://www.youtube.com/watch?v=T05uLRbcloE>
- 2023 Active Inference and The Free Energy Principle with Daniel Friedman**  
 - Video conversation on another channel | No Way Out podcast  
 Source: YouTube <https://www.youtube.com/watch?v=pYleeDb-B5E>
- 2023 Episode 6: Daniel Friedman - Active Inference Institute - What is Active Inference?**  
 - Video conversation on another channel | Denise Holt  
 Source: YouTube <https://www.youtube.com/watch?v=ZyFqRBp1KE0>
- 2022 Artist of Curio Cards 24-26 Shares Incredible Journey through Art and Science**  
 - Video conversation on another channel | Jake Gallen  
 Source: YouTube [https://youtu.be/X\\_F2YFliUnY](https://youtu.be/X_F2YFliUnY)
- 2022 What Ant Behavior Reveals About Complexity Science with Daniel Friedman**  
 - Video conversation on another channel | Making Sense of Complexity  
 Source: YouTube <https://youtu.be/Or98-sG8r0> | YouTube episode <https://www.youtube.com/watch?v=4qsyV38qfTw>
- 2022 What is Active Inference?...with Daniel Friedman**  
 - Video conversation on another channel | Inner Sense  
 Source: YouTube <https://youtu.be/vHnm-zMqLvY>
- 2022 Great discussion with Daniel, OG Curio Card artist, cards 24, 25, 26**  
 - Video conversation on another channel | ZeroG  
 Source: YouTube <https://youtu.be/AHOGPsgIWmU>



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**2022 Daniel Friedman Interview - Adventures in NFTs #3**

- Video conversation on another channel | World Crypto Network

Source: YouTube [https://www.youtube.com/watch?v=mU\\_lwFsEpNg](https://www.youtube.com/watch?v=mU_lwFsEpNg)

**2022 Banging the Drum with Daniel Friedman**

- Video conversation on another channel | Nicholas Pacheco

Source: YouTube <https://www.youtube.com/watch?v=fBuS4yItitw>

**2022 KB7 Fireside: Scaling Principled Games**

- Video conversation on another channel | Kernel

Source: YouTube <https://www.youtube.com/watch?v=vFQk8UR-UVO>

**2022 KBCast Episode 7: Kernel Block 5 Week 6 and special guest Daniel Ari Friedman**

- Video conversation on another channel | Active Inference Lab

Source: YouTube <https://www.youtube.com/watch?v=R5zdTez8CPk>

**2021 Daniel Friedman Interview - Adventures in NFTs #3**

- Video conversation on another channel | World Crypto Network

Source: YouTube [https://youtu.be/mU\\_lwFsEpNg](https://youtu.be/mU_lwFsEpNg)

**2020 Interview with Monica Kang about Complexity Weekend, with co-founder Shaun Swanson and organizer JP Gonzales**

- Video conversation on another channel

Source: YouTube <https://youtu.be/nEaG4xi6MCE>

**2021 Blurb in Science magazine inventing the word disinforme**

- Blurb

Source: Science <https://www.science.org/doi/10.1126/science.abq0904>

**2021 Complexity Weekend featured on InnovatorsBox blog, with quotes from Friedman and others**

- Blurb

Source: InnovatorsBox <https://blog.innovatorsbox.com/complexity-weekend/>

**2020 Blurb in Science magazine about rethinking time use for scientists**

- Blurb

Source: Science <https://www.science.org/>

**2016 Blurb in Science magazine about what ants can teach human city planners**

- Blurb

Source: Science <https://www.science.org/doi/abs/10.1126/science.aag1520>

**2013 Blurb in Science magazine about time travel**

- Blurb

Source: Science <https://www.science.org/>

**2017 Public lecture: Red Vic lecture series, San Francisco. History and philosophy of Art and Biology.**

- Lecture

Source: verification <https://danielarifriedman.com/resume/verify.html>

**2016 Public lecture: Red Vic lecture series, San Francisco. History and philosophy of self-organization, stigmergy, and eusocial insects.**

- Lecture

Source: verification <https://danielarifriedman.com/resume/verify.html>

**2021 Quoted in Dopamine Nation by Dr. Anna Lembke, p. 75: The world is sensory rich and causal poor.**

- Quote

Source: Dopamine Nation <https://www.penguinrandomhouse.com/books/624957/dopamine-nation-by-anna-lembke-md/>

**2025 No Signal 006: Daniel Ari Friedman and Jakub Smekal || Active Inference**

- Profile or Interview

Source: No Signal <https://kernel0x.substack.com/p/no-signal-006-daniel-ari-friedman>

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**2022 Daniel Friedman: Drawing Decentralized Beauty From Biology**

- Profile or Interview | Curio DAO

Source: Curio DAO Mirror [https://mirror.xyz/0x8Fbf5c7Ca1295e892cC865500A8420C3316b889c/ztvis\\_d2IhlBUAb0LI4mCilrvomZr939O1T71a5-R1Y](https://mirror.xyz/0x8Fbf5c7Ca1295e892cC865500A8420C3316b889c/ztvis_d2IhlBUAb0LI4mCilrvomZr939O1T71a5-R1Y)

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**2021 Written interview with RosyBVM**

- Profile or Interview

Source: Rosy Conversation <https://rosybvm.com/2021/01/20/rosy-conversation-with-daniel-ari-friedman/>

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**2019 Radio Interview, KPOO 89.5 FM in San Francisco**

- Radio

Source: KPOO <https://kpoa.com/>

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## Professional Participation, Engagement, and Service (27)

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**2022-2024 ISSS | Board of Directors**

- Vice-President for Communication and Systems Education on the International Society for the Systems Sciences Board.

Source: ISSS <https://www.issss.org/home/>

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**2021-2023 JoVE | Academia**

- Co-editor, Field and Laboratory Methods for Modern Entomology.

Source: JoVE methods collection <https://www.jove.com/methods-collections/567/field-and-laboratory-methods-for-modern-entomology>

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**2021 Cybernetics Society | Membership**

- Member (MCyBS), Cybernetics Society.

Source: Cybernetics Society <http://cybsoc.org/>

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**2020-ongoing IPA | Membership**

- Information Professional Association.

Source: IPA <https://information-professionals.org/>

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**2020-ongoing Active Inference Institute | Community**

- Active Inference Institute co-founder.

Source: Active Inference Institute <https://www.activeinference.institute/>

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**2020-2023 Davis Complexity Group | Community**

- Davis Complexity Group.

Source: verification <https://danielarifriedman.com/resume/verify.html>

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**2019-2024 Complexity Adventures | Community**

- Complexity Adventures co-founder.

Source: Complexity Adventures <https://www.complexityadventures.com/>

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**2018-2021 ALIUS | Membership**

- ALIUS Research Group contributor, editor, and co-organizer.

Source: ALIUS Research Group <https://www.aliusresearch.org/>

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**2018-2019 Wonderfest | Community**

- Wonderfest Science Envoy; science outreach and community training.

Source: Wonderfest <https://wonderfest.org/>

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**2017-2021 ESA | Membership**

- Entomological Society of America.

Source: Entomological Society of America <https://www.entsoc.org/>

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**2017-2019 Stanford | Academia**

- NeuroChoice group, Stanford Neurosciences Institute.

Source: NeuroChoice <https://neurochoice.stanford.edu/purpose>

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**2017-2018 Stanford | Membership**

- Graduate student member of Arts Committee for Stanford Biology department.

Source: Stanford Biology <https://biology.stanford.edu/>

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**2016-2019 Stanford Complexity Group | Community**

- Stanford Complexity Group.

Source: Stanford <https://www.stanford.edu/> | verification <https://danielarifriedman.com/resume/verify.html>

**2016 Stanford | Community**

- Science Teaching Through Art (STAR) outreach program; poster presentations and conversations about ants at local middle and high schools.

Source: Stanford <https://www.stanford.edu/>

**2015 Santa Fe Institute | Workshop**

- Santa Fe Institute Complex Systems Summer School, one-month residential program.

Source: Santa Fe Institute CSSS <https://www.santafe.edu/engage/learn/programs/sfi-complex-systems-summer-school>

**2014-2018 Gordon Lab | Field work**

- Five summers of ant fieldwork during PhD under Prof. Deborah Gordon at Southwestern Research Station, Arizona.

Source: Southwestern Research Station <https://www.amnh.org/research/southwestern-research-station>

**2014 Stanford | Field work**

- Summer field ecology class with Stanford Prof. Rodolfo Dirzo at Las Tuxtlas tropical research station, Mexico.

Source: Los Tuxtlas Biological Station <https://www.ibiologia.unam.mx/eblt/>

**2014 NIMBioS | Workshop**

- Summer workshop on Evolutionary and Quantitative Genetics at NIMBioS, University of Tennessee, Knoxville.

Source: NIMBioS [http://www.nimbios.org/tutorials/TT\\_eqq](http://www.nimbios.org/tutorials/TT_eqq)

**2014 Science | Field work**

- Drosophila work with Dr. Michael Lang in Arizona.

Source: Drosophila Evolution <https://droseu.net/drosophila-evolution/>

**2014-ongoing Science | Academia**

- Invited peer reviewer for 1-10 articles per year across different journals.

Source: Science <https://www.science.org/>

**2014-2023 IUSSI | Membership**

- International Union for the Study of Social Insects.

Source: IUSSI <https://ius.si.org/>

**2014-2017 Stanford | Community**

- Stanford Center for Evolutionary and Human Genomics outreach programs, including teaching genetics in local middle schools and hosting high school students at Stanford.

Source: Stanford Center for Evolution and Human Genomics <https://cehg.stanford.edu/>

**2014-2017 Stanford | Community**

- Stanford Bio Core Explorations hands-on lecture/workshop for undergraduates: Art and Biology.

Source: Stanford Biology <https://biology.stanford.edu/>

**2013 UC Davis | Community**

- Science mentoring at Woodland High School.

Source: Woodland High School <https://whs.wiusd.org/>

**2012 UC Davis | Community**

- Drums and percussion for UC Davis production of RENT.

Source: UC Davis <https://www.ucdavis.edu/>

**2012-2013 UC Davis | Community**

- Volunteer in emergency room at hospital in Sacramento, four hours per week for six months.

Source: UC Davis Health <https://health.ucdavis.edu/>

**2011-2014 UC Davis | Community**

- Chess Club at UC Davis President (2012-2014), organizing free outdoor chess at the farmer's market twice weekly for two years.

Source: UC Davis Campus Recreation <https://campusrecreation.ucdavis.edu/>



## Art Portfolio and Uses (14)

### All Art portfolio

- Posting drawings to an online portfolio since 2009.

Source: Flickr [http://flickr.com/photos/daniel\\_friedman/](http://flickr.com/photos/daniel_friedman/)

### 2021 Curio Cards

- A full Curio Cards set sold at Christie's on October 1, 2021 for 393 ETH, approximately \$1.2 million.

Source: Christie's Curio Cards lot <https://www.christies.com/lot/lot-6337619>

### 2021 Curio Cards

- Curio Cards collection stored in the Arctic World Archive with art and provenance data intended for long-duration preservation.

Source: Top Dog Beach Club <https://topdogbeachclub.com/non-fungible-vault/>

### 2021 Curio Cards

- Public auction and market history for Curio Cards includes major 2021 sales; verify venue-specific claims before reusing exact auction-house wording.

Source: Sotheby's Curio Cards <https://metaverse.sothebys.com/natively-digital/lots/curio-cards-complete-set-1-30-plus-17b>

### 2019 Art used

- Art used in the eBook Rebellng with Care: Exploring open technologies for commoning healthcare.

Source: ResearchGate [https://www.researchgate.net/publication/335911369\\_Rebellng\\_with\\_Care\\_Exploring\\_open\\_technologies\\_for\\_commoning\\_healthcare](https://www.researchgate.net/publication/335911369_Rebellng_with_Care_Exploring_open_technologies_for_commoning_healthcare)

### 2019 Art used

- Art used in Slate.fr blog.

Source: Slate France <https://www.slate.fr/story/173103/fecondation-spermatozoide-ovule-chevalier-belle-au-bois-dormant-mythe-sexisme-naturalisation-biologie?>

### 2019 Art used

- Art used in degrowth blog.

Source: Degrowth.info <https://www.degrowth.info/blog/on-strategies-for-socioecological-transformation>

### 2018 Art used

- Art used in Medium post by Lorena Cabeza, August 23, 2018.

Source: Medium <https://medium.com/@lorenacabeza/manifiesto-la-escritura-y-la-vida-10d4c254c0bf>

### 2018 Art used

- Cover image of Probe magazine.

Source: Probe Magazine <https://www.ssbprobe.com/decentraldogma-danielfriedman>

### 2017 Curio Cards

- Early Ethereum art NFT project; art for Curio Cards 24, 25, and 26.

Source: Curio Cards artist <https://curio.cards/artist/danielfriedman/> | Curio Cards docs <https://docs.curio.cards/the-artists/daniel-friedman>

### 2017 Art used

- Art used in Bas Haring blog.

Source: Me Judice <https://www.mejudice.nl/artikelen/detail/mijn-economiemuseum-de-markt>

### 2017 Art used

- Art used in Vice Motherboard blog.

Source: VICE Motherboard <https://www.vice.com/en/article/ipagk8/a-dollar25-million-book-heist-proves-the-printed-page-is-still-sacred>

### 2016 Art used

- Featured in Genetics Society of America art blog.

Source: Genes to Genomes <https://genestogenomes.org/gsa-art-daniel-friedman/>

### 2016 Art used

- Art featured by Stanford Art of Neuroscience competition.

Source: Stanford Scope <https://scopeblog.stanford.edu/2016/07/06/art-of-neuroscience-competition-highlights-beauty-of-the-brain/>

